STAT 560/561 Lecture Notes

Statistical Inference

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1 Review of Probability

1.1 Probability Measures

We denote $(\Omega, \mathcal{F}, \mathbb{P})$ as the **probability space**, where

- Ω is the **sample space**, a set containing all possible outcome of a certain event;
- $\mathcal{F}$ is the **σ-algebra** on Ω. In particular, it satisfies the following:
  - $\Omega \in \mathcal{F}$;
  - If $A \in \mathcal{F}$, then $A^C = \Omega \setminus A \in \mathcal{F}$;
  - If $\{A_n\}_{n=1}^{\infty} \subset \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

  We denote $\sigma(\Omega)$ as the σ-algebra generated by Ω, which is the smallest σ-algebra that contains Ω. Later, when the sample space is $\mathbb{R}$, we will denote $\mathfrak{B}(\mathbb{R})$ as the **Borel σ-algebra** on $\mathbb{R}$, which is generated by the collection of open sets in $\mathbb{R}$.

  In the event of a coin-tossing, if we denote H to be the head, and T to be the tail, then the sample space is $\Omega = \{H, T\}$, and the σ-algebra on Ω is $\mathcal{F} := \{\varnothing, \{H, T\}, \{H\}, \{T\}\}$.
- $\mathbb{P}$ is the **probability measure** on Ω, which is a mapping $\mathbb{P} : \mathcal{F} \to [0, 1]$ such that it satisfies *Kolmogorov's axioms*:
  - $\mathbb{P}(\Omega) = 1$;
  - $\mathbb{P}(A) \geq 0, \forall A \in \mathcal{F}$;
  - If $A_i \cap A_j = \varnothing, \forall i \neq j$, then $\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(A_n)$.

  Based on Kolmogorov's axioms more properties of probability measure follow:
  - $\mathbb{P}(A^C) = \mathbb{P}(\Omega \setminus A) = 1 - \mathbb{P}(A)$;
  - If $A \cap B = \varnothing$, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$;
  - If $A \subset B$, then $\mathbb{P}(A) \subset \mathbb{P}(B)$;
  - **The principle of inclusion and exclusion:** Let $A_1, A_2, \cdots, A_n \in \mathcal{F}$, then

$$\mathbb{P}\left(\bigcup_{k=1}^{n} A_k\right) = \sum_{k=1}^{n} \mathbb{P}(A_k) - \sum_{k_1<k_2} \mathbb{P}(A_{k_1} \cap A_{k_2}) + \cdots + (-1)^{n+1} \sum_{k_1<\cdots<k_n} \mathbb{P}(A_{k_1} \cap \cdots \cap A_{k_n}).$$

  In the special case of $n = 2, 3$, we have the following: $\forall A, B, C \in \Omega$,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B);$$

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).$$

Theorem 1 (**Continuity of $\mathbb{P}$**). Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of non-decreasing events in $\mathcal{F}$, i.e $A_n \subset A_{n+1}$, then

$$\mathbb{P}(\lim_{n\to\infty} A_n) = \lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right).$$

Proof. Let $B_1 = A_1, B_2 = A_2 \setminus A_1, \cdots, B_n = A_n \setminus \bigcup_{i=1}^{n-1} A_i$. Then $B_i \cap B_j = \varnothing, \forall i \neq j$, and $A_n = \bigcup_{i=1}^{n} B_i$. Then by countable additivity of $\mathbb{P}$, we have

$$\mathbb{P}(A_n) = \mathbb{P}\left(\bigcup_{i=1}^{n} B_i\right) = \sum_{i=1}^{n} \mathbb{P}(B_i).$$

By taking $n \to \infty$, we have

$$\lim_{n\to\infty} \mathbb{P}(A_n) = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbb{P}(B_i) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} B_n\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}(\lim_{n\to\infty} A_n).$$

□

Theorem 2 (**Continuity of $\mathbb{P}$, cont'd**). Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of non-increasing events in $\mathcal{F}$, i.e $A_{n+1} \subset A_n$, then

$$\mathbb{P}(\lim_{n\to\infty} A_n) = \lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right).$$

Proof. Consider $\{A_n^C\}_{n=1}^{\infty}$ which is non-decreasing. □

We next review conditional probability.

Definition 1. Suppose $A, H \in \Omega$, then we define $\mathbb{P}(A|H)$ to be the probability of A *given that* H occurs, known as the **conditional probability** of A given H.

The conditional probability of A given H is given by

$$\mathbb{P}(A|H) = \frac{\mathbb{P}(A \cap H)}{\mathbb{P}(H)}.$$

We say A, H are **independent** if $\mathbb{P}(A \cap H) = \mathbb{P}(A) \cap \mathbb{P}(H)$. Using the above formula, we have that $\mathbb{P}(A|H) = \mathbb{P}(A)$, i.e, the occurrence of H does not influence the occurrence of A.

Note that conditional probability on H, $\mathbb{P}(\cdot|H)$ is also a probability measure on $(H, \sigma(H))$, i.e, it satisfies Kolmogorov's axioms:

- $\mathbb{P}(H|H) = 1$;
- $\mathbb{P}(B|H) \geq 0, \forall B \in \sigma(H)$;
- $\mathbb{P}(B|H) = 1 - \mathbb{P}(B^C|H)$, where $B^C = H \setminus B$;
- If $H_1, H_2, \cdots$ are disjoint, then $\mathbb{P}\left(\bigcup_{n=1}^{\infty} H_n \middle| B\right) = \sum_{n=1}^{\infty} \mathbb{P}(H_n|B)$.

A sequence $\{A_n\}_{n=1}^{\infty}$ of events in $\mathcal{F}$ is called a **partition** of Ω if $A_i \cap A_j = \varnothing, i \neq j$ and $\bigcup_{n=1}^{\infty} A_n = \Omega$.

Theorem 3 (**Law of Total Probability**). Let $\{A_n\}_{n=1}^{\infty}$ be a partition of Ω, then $\forall B \in \mathcal{F}$,

$$\mathbb{P}(B) = \sum_{n=1}^{\infty} \mathbb{P}(B|A_n)\mathbb{P}(A_n).$$

Proof. We know that $B = B \cap \Omega = B \cap \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} B \cap A_n$, thus

$$\begin{aligned}\mathbb{P}(B) &= \mathbb{P}\left(\bigcup_{n=1}^{\infty} B \cap A_n\right) \\ &= \sum_{n=1}^{\infty} \mathbb{P}(B \cap A_n), \quad \textit{since } B \cap A_n\textit{'s are disjoint} \\ &= \sum_{n=1}^{\infty} \mathbb{P}(B|A_n)\mathbb{P}(A_n).\end{aligned}$$

□

Theorem 4 (**Bayes Theorem**). Let $\{A_n\}_{n=1}^{\infty}$ be a partition of Ω, with $\mathbb{P}(A_n) > 0, \forall n$. Then $\forall B \in \mathcal{F}, \mathbb{P}(B) > 0$, and $\forall k \in \mathbb{N}$,

$$\mathbb{P}(A_k|B) = \frac{\mathbb{P}(A_k)P(B|A_k)}{\sum_{n=1}^{\infty} P(A_n)\mathbb{P}(B|A_n)}.$$

Proof. We know that $\mathbb{P}(A_k|B) = \frac{\mathbb{P}(A_k \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A_k)\mathbb{P}(A_k)}{\mathbb{P}(B)}$, and use law of total probability to represent $\mathbb{P}(B)$. □

1.2 Random Variable and Distributions

Definition 2 (**Random variable**). A random variable X on $(\Omega, \mathcal{F})$ is a mapping (measurable function) $X : \Omega \to \mathbb{R}$, such that for any Borel set $B \in \mathfrak{B}(\mathbb{R})$,

$$X^{-1}(B) := \{w \in \Omega : X(\omega) \in B\} \in \mathcal{F}.$$

Some analysis show that, the Borel σ-algebra is also generated by the intervals of the form $(a, b]$, hence if $\forall x \in \mathbb{R}$,

$$X^{-1}((-\infty, x]) = \{\omega \in \Omega : X(\omega) \in (-\infty, x]\} \in \mathcal{F},$$

then X is a random variable.

Example 1 (Coin Tossing). Consider tossing a fair coin two times, denote H = head face up and T = tail face up, then we know that $\Omega = \{HH, HT, TH, TT\}$ and $\mathcal{F}$ is the σ-algebra generated by Ω. We define $X(\omega) =$ Number of H's in ω, then clearly $X(HH) = 2, X(HT) = X(TH) = 1, X(TT) = 0$, and further we have

$$X^{-1}((-\infty, x]) = \begin{cases} \varnothing : x < 0 \\ \{TT\} : 0 \le x < 1 \\ \{TT, HT, TH\} : 1 \le x < 2 \\ \Omega : 2 \le x \end{cases} \in \mathcal{F}$$

and thus X is a random variable.

We mainly focus on two types of random variables: **discrete random variables** and **continuous random variables**. For discrete random variables X, it takes on countably many values (for example, $\mathbb{N}$);

Definition 3 (**CDF**)**.** For a random variable X, the **cumulative distribution function (CDF)** of X, denoted as $F_X(\cdot)$, is defined by $F_X(x) = \mathbb{P}(X \leq x)$, $x \in \mathbb{R}$.

Based on the definition, we know the following about CDF:

- $F_X(\cdot)$ is non-negative and non-decreasing;
- $F_X(-\infty) = 0, F_X(\infty) = 1$;
- $F_X(\cdot)$ is right continuous.

Definition 4 (**pmf**)**.** For a discrete random variable X, the **probability mass function (pmf)** of X, denote as $f_X(\cdot)$, is defined by $f_X(X = x) = \mathbb{P}(X = x)$;

Definition 5 (**continuous random variable, pdf**)**.** A random variable X is said to be continuous if there exists a non-negative function $f_X(\cdot)$, such that $\int_{\mathcal{S}} f_X(x)dx = 1$ and $\mathbb{P}(X \in B) = \int_B f_X(x)dx$, where $\mathcal{S}$ is the support of X, $B \in \mathfrak{B}(\mathbb{R})$. Then $f_X(\cdot)$ is the **probability density function (pdf)** of X, and

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt,$$

when $F_X(\cdot)$ is differentiable at x, we have $F'_X(x) = f_X(x)$.

Example 2 (**examples of random variables**)**.** Here are some examples of commonly used random variables:

- **Discrete random variables:**
  - Bernoulli random variable $X \sim \text{Bern}(p)$: Where X has binary outcomes: $X = \{0, 1\}$, $\mathbb{P}(X = 1) = p, \mathbb{P}(X = 0) = 1 - p$. A single coin toss can be modeled by a Bernoulli random variable;
  - Binomial random variable $X \sim \text{Bino}(n, p)$: Where X can be viewed as a sum of n independent Bernoulli random variables $Y_i \sim \text{Ber}(p)$. Then we have

    $$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \cdots, n.$$

    If we toss a coin n times and record the number of heads, then it would follow a Binomial distribution.
  - Poisson random variable $X \sim \text{Pois}(\lambda)$, where X can be viewed as a "modified Binomial" distribution but n is large and p is small. The probability mass function is given by

    $$\mathbb{P}(X = k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!}, \quad k = 0, 1, \cdots$$

- **Continuous random variables:**

– Normal random variable $X \sim N(\mu, \sigma^2)$: One of the most commonly used continuous random variables. Its probability density function is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right);$$

– Exponential random variable $X \sim \text{Exp}(\lambda)$: Its probability density function is given by

$$f_X(x) = \frac{1}{\lambda} e^{-\lambda x}.$$

Theorem 5 (**Transformation of random variable**)**.** If X is a random variable, and $g(\cdot)$ is a Borel measurable function (i.e, $\forall x \in \mathbb{R}, g^{-1}((-\infty, x]) \in \mathfrak{B}(\mathbb{R})$), then $g(X)$ is again a random variable.

We may then study the transformations of random variables. For notation simplicity, we only focus on continuous random variables. Let X be a continuous random variable and $g(\cdot)$ be a measurable function, then the CDF of $g(X)$ can be obtained by

$$F_{g(X)}(x) = \mathbb{P}(g(X) \leq x) := \mathbb{P}(X \leq g^{-1}(x)) := \int_B f_X(t)dt,$$

where $B := \{t \in \mathbb{R}, t \leq g^{-1}(x)\}$. If differentiable, the cdf can be obtained by differentiating the CDF and apply the fundamental theorem of calculus.

1.3 Moments

Definition 6. Let X be a random variable, we define the **expectation** of X, denote as $\mathbb{E}[X]$, or μ_X, as follow:

- If X is a discrete random variable taking on countably many values from I, then

$$\mathbb{E}[X] = \sum_{i \in I} i \cdot \mathbb{P}(X = i).$$

- If X is a continuous random variable, denote its support by S, the pdf by $f_X(x)$, then

$$\mathbb{E}[X] = \int_S x f_X(x) dx.$$

Expectation satisfies *linearity*. That is, if $X_1, \cdots, X_n$ are random variables and $a_1, \cdots, a_n \in \mathbb{R}$, then

$$\mathbb{E}\left[\sum_{i=1}^n a_i X_i\right] = \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

For a measurable function $g(\cdot)$, if $\mathbb{E}[g(X)]$ exists, then:

- If X is a discrete random variable taking on countably many values from I, then

$$\mathbb{E}[g(X)] = \sum_{i \in I} g(i) \cdot \mathbb{P}(X = i);$$

- If X is a continuous random variable with support S and pdf $f_X(x)$, then

$$\mathbb{E}[f(X)] = \int_S g(x) \cdot f_X(x) dx.$$

Definition 7. Let X be a random variable, the **kth central moment** of X, is given by $\mathbb{E}[(X - \mathbb{E}[X])^k]$. When $k = 2$, $\mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ is also denoted as the **variance** of X, denote as $\mathbb{V}\mathbf{ar}(X)$.

Given $a, b \in \mathbb{R}$, we have that $\mathbb{V}\mathbf{ar}(aX + b) = a^2\mathbb{V}\mathbf{ar}(X)$. We next discuss some inequalities related to moments.

Theorem 6 (Markov's inequality). Let X be a non-negative random variable and $\mathbb{E}|X| < \infty$, then $\forall t > 0$,

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}.$$

Proof.

$$\mathbb{E}[X] := \int_{-\infty}^{t} x f_X(x) dx + \int_t^{\infty} x f_X(x) dx \geq \int_t^{\infty} t f_X(x) dx = t\mathbb{P}(X \geq t).$$

□

In general, if a measurable function $g(\cdot)$ is monotonically increasing and $g(t) \neq 0$, then

$$\mathbb{P}(X \geq t) = \mathbb{P}(g(X) \geq g(t)) \leq \frac{\mathbb{E}[g(X)]}{g(t)}.$$

Theorem 7 (Chebyshev's inequality). Let X be a random variable, such that $\mu = \mathbb{E}[X], \mathbb{E}[X^2] < \infty$, then $\forall \epsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \epsilon) \leq \frac{\mathbb{V}\mathbf{ar}(X)}{\epsilon^2}.$$

Proof. Since $g(x) = x^2$ is monotonically increasing on $\mathbb{R}^+$, then

$$\mathbb{P}(|X - \mu| \geq \epsilon) = \mathbb{P}((X - \mu)^2 \geq \epsilon^2).$$

Then one can apply Markov's inequality and use the fact that $\mathbb{V}\mathbf{ar}(X) = \mathbb{E}[(X - \mu)^2]$. □

Definition 8 (MGF). Let X be a random variable, the **moment generating function (mgf)** of X, denote as $M_X(t), t \in \mathbb{R}$ (if exists), is defined by $M_X(t) = \mathbb{E}[e^{tX}]$.

The reason why it is called moment generating function is because the moments of X ($\mathbb{E}[X^k]$), can be obtained by differentiating $M_X(t)$ with respect to t. Note that

$$\frac{d}{dt} M_X(t)\Big|_{t=0} := \mathbb{E}[Xe^{tX}]\Big|_{t=0} = \mathbb{E}[X],$$

$$\frac{d^2}{dt^2} M_X(t)\Big|_{t=0} := \mathbb{E}[X^2 e^{tX}]\Big|_{t=0} = \mathbb{E}[X^2].$$

Theorem 8 (**Chernoff's bound**). Let X be a random variable, then for any $a \in \mathbb{R}$,

$$\mathbb{P}(X \geq a) \leq \inf_{t>0} e^{-at} M_X(t).$$

Proof. Since $g(x) = e^{xt}$ is increasing for $t > 0$, hence

$$\mathbb{P}(X \geq a) = \mathbb{P}(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}} = e^{-ta} \cdot M_X(t), \forall t > 0.$$

□

We may also derive the Chernoff's bound for the other tail: Suppose $t < 0$, then $g(x) = e^{tx}$ is decreasing, hence

$$\mathbb{P}(X \leq a) = \mathbb{P}(e^{tX} \geq e^{ta}) \leq \frac{\mathbb{E}[e^{tX}]}{e^{ta}} = e^{-at} \cdot M_X(t), t < 0,$$

by taking the infinmum, we have

$$\mathbb{P}(X \leq a) \leq \inf_{t<0} e^{-at} M_X(t).$$

Theorem 9 (**Hoeffding's inequality**). Let $X_1, \cdots, X_n$ be independent, and $\mathbb{E}[X_i] = 0, a_i \leq X_i \leq b_i$ almost surely, then $\forall \epsilon > 0$, and $\forall t$,

$$\mathbb{P}\left(\sum_{i=1}^{n} X_i > t\right) \leq \exp\left(-\epsilon t + \frac{\epsilon^2}{8} \sum_{i=1}^{n} (b_i - a_i)^2\right).$$

Note that MGF uniquely determines a random variable, however not all random variables have MGF. For example, the Cauchy random variable $X \sim \text{Cauchy}(0, 1)$, with pdf

$$f_X(x) = \frac{1}{\pi(1 + x^2)}, \quad x \in \mathbb{R}.$$

Then the MGF does not exist:

$$M_X(t) = \mathbb{E}[e^{tX}] = \int_{\mathbb{R}} \frac{e^{tx}}{\pi(1 + x^2)} dx = \infty.$$

To avoid this issue, we sometime use the **characteristic function** of a random variable X defined by

$$\varphi_X(t) = \mathbb{E}[e^{iXt}], \quad t \in \mathbb{R}.$$

Characteristic function for a random variable X always exists, and it uniquely determines X. Again take $X \sim \text{Cauchy}(0, 1)$, then for $t > 0$,

$$\varphi_X(t) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{e^{itx}}{1 + x^2} dx = \frac{1}{\pi} \lim_{R \to \infty} \int_{\gamma} \frac{e^{itz}}{1 + z^2} dz - \int_0^{\pi} \frac{i\theta R e^{i\theta} e^{itRe^{i\theta}}}{1 + R^2 e^{2i\theta}} d\theta = e^{-t},$$

where the first integral can be easily computed by residue theorem (γ is the upper circle with radius R).

Theorem 10 (**Jensen's inequality**). Let X be a random variable, and $\mathbb{E}|X| < \infty$.

- If g is a convex function (i.e $g''(x) > 0$), then $g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)]$;
- If g is a concave function (i.e $g''(x) < 0$), then $g(\mathbb{E}[X]) \geq \mathbb{E}[g(X)]$.

Proof. Let $y = ax + b$ be the tangent line to $g(x)$ at $x_0 = \mathbb{E}[X]$. If g is convex, then the graph of $g(x)$ is below y, i.e., $g(x) \leq y$, hence

$$\mathbb{E}[g(X)] \leq \mathbb{E}[aX + b] = y(\mathbb{E}[X]) = g(\mathbb{E}[X]).$$

The argument for g is concave is similar. □

1.4 Multiple Random Variables

Definition 9 (**Joint distribution**). Let $X_1, \cdots, X_n$ be n random variables.

- $(X_1, \cdots, X_n)$ is said to be discrete if they take on countably many values. Then the cumulative distribution function of $(X_1, \cdots, X_n)$ is given by

$$F(x_1, \cdots, x_n) = \sum_{(a_1, \cdots, a_n) \in S} \mathbb{P}(X_1 = a_1, \cdots, X_n = a_n),$$

  where $B := \{(a_1, \cdots, a_n) : x_1 \leq a_1, \cdots, x_n \leq a_n\}$.

- $(X_1, \cdots, X_n)$ is said to be continuous if there exists a non-negative function $f(x_1, \cdots, x_n)$, such that $\int_S f(x_1, \cdots, x_n) d\boldsymbol{x} = 1$ and

$$\mathbb{P}(X_1 \leq x_1, \cdots, X_n \leq x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(t_1, \cdots, t_n) dt_n \cdots dt_1.$$

Similar to the definition we had before, random variables X, Y are **independent** if $\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A) \cdot \mathbb{P}(Y \in B)$ for any $A, B \in \mathfrak{B}(\mathbb{R})$. Under this case, if f_X is the density for X and f_Y is the density for Y, denote $f_{X,Y}$ as the joint density of X, Y, then $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y), \forall x, y \in \mathbb{R}$.

If X, Y are not independent, we may define the **covariance** of X, Y, denote as $\mathbb{C}\mathbf{ov}(X, Y)$, as

$$\mathbb{C}\mathbf{ov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y],$$

where

$$\mathbb{E}[XY] = \iint_{\mathbb{R}^2} xy f_{X,Y}(x, y) dy dx.$$

Some properties of covariance follow:

- If X, Y are independent, then $\mathbb{C}\mathbf{ov}(X, Y) = 0$;
- $\mathbb{C}\mathbf{ov}(aX, Y) = \mathbb{C}\mathbf{ov}(X, aY) = a\mathbb{C}\mathbf{ov}(X, Y)$;
- $\mathbb{C}\mathbf{ov}\left(\sum_{i=1}^{n} X_i, \sum_{j=1}^{m} Y_j\right) = \sum_{i,j} \mathbb{C}\mathbf{ov}(X_i, Y_j)$;

- $\mathbb{V}\mathbf{ar}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathbb{V}\mathbf{ar}(X_i) + 2\sum_{i\neq j} \mathbb{C}\mathbf{ov}(X_i, X_j)$.

The **correlation** of two random variables X, Y, denote as $\rho(x, y)$, is defined by

$$\rho(X, Y) = \frac{\mathbb{C}\mathbf{ov}(X, Y)}{\sqrt{\mathbb{V}\mathbf{ar}(X) \cdot \mathbb{V}\mathbf{ar}(Y)}},$$

it can be shown that $\rho(X, Y) \in [-1, 1]$.

FUN FACT: EXPECTATION IS AN INNER PRODUCT.

Consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, define $L^2 := L^2(\Omega, \mathcal{F}, \mathbb{P})$ as the space of square-integrable random variables with

$$\mathbb{E}[X^2] = \int_\Omega x^2 f_X(x) dx,$$

we claim that L^2 defines a Hilbert space (a complete inner product space), with inner product defined by $\langle X, Y\rangle := \mathbb{E}[XY]$ and $\langle X, Y\rangle = 0$ if and only if X, Y are independent (or called orthogonal). The norm equipped on L^2 is simply defined by $||X||_{L^2} := \mathbb{E}[X^2]^{1/2}$.

By viewing expectation as a L^p space, we can then derive inequalities with respect to $\mathbb{E}|X|^p$ using L^p theory. Below, we assume X, Y come from the same probability space.

Theorem 11 (**Cauchy-Bunyakovsky-Schwarz inequality**)**.** Let X, Y be random variables, then

$$(\mathbb{E}|XY|)^2 \leq \mathbb{E}|X|^2 \mathbb{E}|Y|^2.$$

Theorem 12 (**Hölder's inequality**)**.** Let X, Y be random variables, $p, q > 1$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$\mathbb{E}|XY| \leq (\mathbb{E}|X|^p)^{1/p} \cdot (\mathbb{E}|Y|^q)^{1/q}.$$

Theorem 13 (**Minkouski's inequality**)**.** Let X, Y be random variables, then

$$\mathbb{E}|X + Y|^r \leq C_r(\mathbb{E}|X|^r + \mathbb{E}|Y|^r), \quad C_r = \begin{cases} 1 & r \in [0, 1] \\ 2^{r-1} & r > 1 \end{cases}.$$

I will omit the proof for those three inequalities above.

For random variables $X_1, \cdots, X_n$ with joint distribution $f_{X_1,\cdots,X_n}$, it is also possible to find the distribution of $\boldsymbol{g}(\boldsymbol{X}) = (g_1(X_1), g_2(X_2), \cdots, g_n(X_n))$. This can be done by change of variable formula described below:

Theorem 14 (**Change of variable formula**)**.** Let $\boldsymbol{X} = \begin{pmatrix} X_1 & X_2 & \cdots & X_n \end{pmatrix}^\top$ be an n-dimensional random vector with joint PDF $f_{\boldsymbol{X}}(\boldsymbol{x})$. Let $G \in C^2(\mathbb{R}^n) : \mathbb{R}^n \to \mathbb{R}^n$ be invertible and let $H = G^{-1}$.

Suppose a random vector $\boldsymbol{Y} = \begin{pmatrix} Y_1 & Y_2 & \cdots & Y_n \end{pmatrix}^\top$ is given by $\boldsymbol{Y} = G(\boldsymbol{X})$, hence $\boldsymbol{X} = H(\boldsymbol{Y})$, and

$$\boldsymbol{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix} = \begin{pmatrix} H_1(Y_1, Y_2, \cdots, Y_n) \\ H_2(Y_1, Y_2, \cdots, Y_n) \\ \vdots \\ H_n(Y_1, Y_2, \cdots, Y_n) \end{pmatrix},$$

then the PDF of $\boldsymbol{Y}$ is given by

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = f_{\boldsymbol{X}}(H(\boldsymbol{y})) \cdot |J|$$

where J is the Jacobian of H defined by $J = \left(\frac{\partial H_i}{\partial y_j}\right)_{i,j} \in M_n(\mathbb{R})$.

Definition 10 (**Conditional distribution**)**.** Let X, Y be random variables, $f_{X,Y}(\cdot,\cdot)$ be the joint density of X, Y. The **conditional distribution** of X given $Y = y$, denote as $f_{X|Y=y}(x|y)$ is defined by

$$f_{X|Y=y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)},$$

where $f_Y(y)$ is the **marginal distribution** of Y, which is defined by

$$f_Y(y) = \sum_{x \in I} \mathbb{P}(X = x, Y = y), \quad \text{if } X, Y \text{ are discrete;}$$

and

$$f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x,y)dx, \quad \text{if } X, Y \text{ are continuous.}$$

1.5 Convergence of Random Variables

We will study the convergence of sequence of random variables $\{X_n\}_{n=1}^\infty$. We introduce 4 different kinds of convergence:

Definition 11. Let $\{X_n\}_{n=1}^\infty$ be a sequence of random variables, X be another random variable.

- **Weak convergence:** We say $X_n \to X$ weakly (or convergence in distribution, convergence in law), if $\lim_{n\to\infty} \mathbb{P}(X_n \le x) = \mathbb{P}(X \le x)$ for every x such that $x \mapsto \mathbb{P}(X \le x)$ is continuous.
- **Convergence in probability:** We say $X_n \to X$ in probability ($X_n \xrightarrow{P} X$), if $\forall \epsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(\omega : |X_n(\omega) - X(\omega)| > \epsilon) = 0.$$

- **Convergence in L^p:** We say $X_n \to X$ in L^p ($X_n \xrightarrow{L^p} X$), if $\lim_{n\to\infty} \mathbb{E}|X_n - X|^k = 0$. When $p = 2$, we way $X_n \to X$ in quadratic mean.
- **Convergence almost surely:** We say $X_n \to X$ almost surely ($X_n \xrightarrow{a.s} X$), if

$$\mathbb{P}\left(\lim_{n\to\infty} |X_n - X| = 0\right) = 1.$$

Theorem 15. If $X_n \xrightarrow{L^p} X$, then $X_n \xrightarrow{P} X$.

Proof. The proof is straightforward, we have

$$\mathbb{P}\{|X_n - X| > \epsilon\} = \mathbb{P}\{|X_n - X|^p > \epsilon^p\} \leq \frac{\mathbb{E}|X_n - X|^p}{\epsilon^p} \to 0.$$

□

Theorem 16. If $X_n \xrightarrow{P} X$, then there exists a subsequence n_k of $\mathbb{N}$ such that $X_{n_k} \xrightarrow{a.s} X$.

Proof. Assume $X_n \xrightarrow{P} X$, then $\forall k \geq 1$, $\exists n_k$ such that $\mathbb{P}(\{X_{n_k} - X| > 1/k\}) \leq 1/k^2$, denote $A_k := \{|X_{n_k} - X| > 1/k\}$, then by *Borel-Cantelli Lemma*, we have

$$\mathbb{P}\left(\bigcap_{l=1}^{\infty}\bigcup_{k=l}^{\infty} A_k\right) = \lim_{l\to\infty} \mathbb{P}\left(\bigcup_{k=l}^{\infty} A_k\right) \leq \lim_{l\to\infty} \sum_{k=l}^{\infty} \mathbb{P}(A_k) = 0,$$

meaning that for almost everywhere, $\exists l$, such that $\forall k \geq l : |X_{n_k} - X| \leq 1/k$, which means that for almost everywhere, $\lim_{k\to\infty} |X_{n_k} - X| = 0$, thus $X_{n_k} \xrightarrow{a.s} X$. □

The figure below shows the relations between different modes of convergence. The arrows in red means direct implication without any further condition, while arrows in orange will hold if extra conditions are given. The general structure is that, convergence in probability implies convergence almost surely along a subsequence; Convergence in probability with uniform integrability would imply convergence in L^p; Convergence almost surely when dominated convergence theorem applies will imply convergence in L^p.

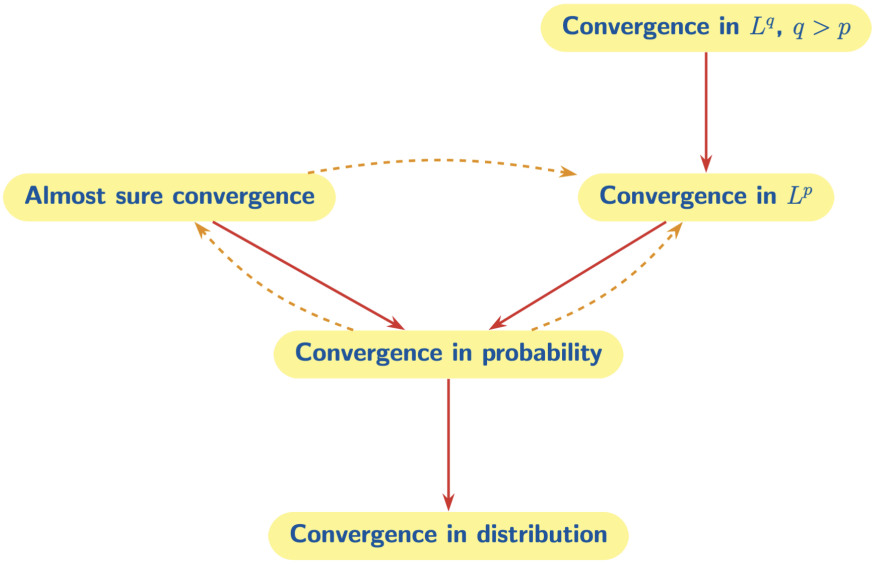


Figure 1: Different modes of convergences and their relations

Theorem 17 (**Continuous mapping theorem**). Let $\{X_n\} \subset \mathbb{R}^k$ be a sequence of random vectors, $X \subset \mathbb{R}^k$ be another random vector. Suppose $g : \mathbb{R}^k \to \mathbb{R}^m$ is continuous almost everywhere on the support of $\{X_n\}, X$, then

$$X_n \to X \implies g(X_n) \to g(X),$$

where the convergence could be convergence in distribution, in probability, or almost everywhere.

Theorem 18 (**Slutsky's theorem**)**.** Let $\{X_n\}, \{Y_n\}, X, Y$ be random vectors, $c \in \mathbb{R}$. Assume $X_n \xrightarrow{d} X, Y_n \xrightarrow{P} c$, then

$$X_n + Y_n \xrightarrow{d} X + c; \quad Y_n X_n \xrightarrow{d} cX; \quad \frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c}, \quad c, Y_n \neq 0.$$

Theorem 19 (**Lévy's continuity theorem**)**.** The characteristic function of X is defined as

$$\mathbb{1}_X(t) := \mathbb{E}[\exp(itX)].$$

Then $X_n \xrightarrow{d} X$ iff $\mathbb{1}_{X_n(t)} \overset{\text{pointwise}}{\to} \mathbb{1}_{X(t)}$ for all $t \in \mathbb{R}$.

Definition 12 (**Stochastic o notations**)**.** .

- A sequence of random vectors $\{X_n\}$ is **bounded in probability** (or uniformly tight), if $\forall \epsilon > 0$, $\exists M_\epsilon$, such that $\sup_n \mathbb{P}(||X_n|| \geq M_\epsilon) \leq \epsilon$. In this case we say $X_n = O_P(1)$.
- If a sequence of random vectors $\{X_n\}$ satisfies $X_n \xrightarrow{P} 0$, then we say $X_n = o_P(1)$.

The o notation is not strictly "equal", it is meant to be read from left to right. For example, the expression

$$o_P(1) + o_P(1) = o_P(1)$$

means: If $X_n \xrightarrow{P} 0$ and $Y_n \xrightarrow{P} 0$, then $X_n + Y_n \xrightarrow{P} 0$. To see this, for any $\epsilon > 0$,

$$\mathbb{P}(|X_n + Y_n| \geq \epsilon) \leq \mathbb{P}(|X_n| \geq \epsilon/2 \vee [1] |Y_n| \geq \epsilon/2) \leq \mathbb{P}(|X_n| \geq \epsilon/2) + \mathbb{P}(|Y_n| \geq \epsilon/2) \to 0,$$

for large n.

Proposition 1. The following hold:

- $o_P(1) \cdot O_P(1) = o_P(1)$;
- $o_P(1) + O_P(1) = O_P(1)$.

Proof. The first "equation" is saying that, if $X_n \xrightarrow{P} 0$ and Y_n is bounded in probability, then $X_n \cdot Y_n \xrightarrow{P} 0$. For any $\epsilon > 0$, by definition $\exists M_\epsilon$ such that $\sup_n \mathbb{P}(|Y_n| \geq M_\epsilon) \leq \epsilon$. Note that

$$\begin{aligned}\mathbb{P}(|X_n Y_n| \geq \epsilon) = \mathbb{P}(|X_n| \cdot |Y_n| \geq \epsilon) &\leq \mathbb{P}\left(|X_n| \geq \frac{\epsilon}{M_\epsilon} \vee |Y_n| \geq M_\epsilon\right) \\ &\leq \mathbb{P}\left(|X_n| \geq \frac{\epsilon}{M_\epsilon}\right) + \mathbb{P}(|Y_n| \geq M_\epsilon).\end{aligned}$$

Since $X_n \xrightarrow{P} 0$. So there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have $\mathbb{P}(|X_n| \geq \epsilon/M_\epsilon) \leq \epsilon$. Hence combine with the equation above, we have

$$\mathbb{P}(|X_n Y_n| \geq \epsilon) \leq 2\epsilon,$$

[1] The symbol "$\vee$" means "or"; In contrast "$\wedge$" means "and".

hence $X_n Y_n \xrightarrow{P} 0$ as desired.

The second "equation" is saying that, if $X_n \xrightarrow{P} 0$ and Y_n is bounded in probability, then $X_n + Y_n$ is bounded in probability. For any $\epsilon > 0$, by definition $\exists M_\epsilon$ such that $\sup_n \mathbb{P}(|Y_n| \geq M_\epsilon) \leq \epsilon$; also $\exists \delta_\epsilon$ such that $\mathbb{P}(|X_n| \geq \delta_\epsilon) \leq \epsilon$ for large n. Note that

$$\mathbb{P}(|X_n + Y_n| \geq M_\epsilon + \delta) \leq \mathbb{P}(|X_n| \geq \delta \vee |Y_n| \geq M_\epsilon) \leq \mathbb{P}(|X_n| \geq \delta_\epsilon) + \mathbb{P}(|Y_n| \geq M_\epsilon) \leq 2\epsilon,$$

so $X_n + Y_n$ is bounded i probability. □

Theorem 20 (**Central limit theorem**). Let $\{X_i\}_{i=1}^{\infty} \overset{\text{i.i.d}}{\sim} f$ and assume $\mathbb{E}[X^2] < \infty$, then

$$\sqrt{n}\Big(\overline{X}_n - \mathbb{E}[X]\Big) \xrightarrow{d} N(0, \mathbb{V}\mathbf{ar}(X)).$$

Proof. WLOG assume $\mathbb{E}X = 0, \mathbb{V}\mathbf{ar}(X) = 1$ then

$$\begin{aligned}
\mathbb{1}_{\sqrt{n}\overline{X}_n}(t) &= \mathbb{E}\Big[\exp\Big(\sqrt{n} i t \overline{X}_n\Big)\Big] \\
&= \mathbb{E}\left[\exp\left(\frac{it(X_1 + \cdots + X_n)}{\sqrt{n}}\right)\right] \\
&= \left(\mathbb{E}\left[\exp\left(\frac{itX}{\sqrt{n}}\right)\right]\right)^n \\
&= \left[\mathbb{1}_X\left(\frac{t}{\sqrt{n}}\right)\right]^n.
\end{aligned}$$

Then a Taylor expansion around 0 will yield

$$\begin{aligned}
\mathbb{1}_X\left(\frac{t}{\sqrt{n}}\right) &= \mathbb{1}_X(0) + \mathbb{1}'_X(0) \cdot \frac{t}{\sqrt{n}} + \mathbb{1}''_X \cdot \frac{t^2}{2n} + o\left(\frac{1}{n}\right) \\
&= 1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)
\end{aligned}$$

This is because

$$\begin{aligned}
\mathbb{1}'_X(t)\Big|_{t=0} &= \frac{d}{dt}\Big|_{t=0} \int_B f(x) \cdot \exp(itX) dx \\
&= \int_B \frac{d}{dt}\Big|_{t=0} f(x) \cdot \exp(itX) dx \\
&= \int_B ixf(x) \cdot \exp(itX)\Big|_{t=0} dx \\
&:= i\mathbb{E}[X] \\
&= 0.
\end{aligned}$$

A similar statement can be drawn: $\mathbb{1}_X''(0) = i^2\mathbb{E}[X^2] = -1$. Use the fact that $\left(1+\frac{x}{n}\right)^n \sim e^x$ for large n, we have

$$\begin{aligned}\mathbb{1}_{\sqrt{n}\overline{X}_n}(t) &= \left[\mathbb{1}_X\left(\frac{t}{\sqrt{n}}\right)\right]^n \\ &= \left[1-\frac{t^2}{2n}+o\left(\frac{1}{n}\right)\right]^n \\ &= \exp\left(-\frac{t^2}{2}\right).\end{aligned}$$

By the uniqueness of the characteristic function, we have finished the proof.

□

We next present a "stronger" central limit theorem:

Theorem 21 (**Lindeberg-Feller CLT**)**.** For each $n \geq 1$, let $Y_{n,1}, Y_{n,2}, \cdots, Y_{n,k_n}$ be independent with finite variance, and $\forall \epsilon > 0$,

$$\sum_{i=1}^{k_n} \mathbb{E}\left[||Y_{n,i}||^2 \cdot \mathbb{1}_{||Y_{n,i}||>\epsilon}\right] \to 0, \quad \sum_{i=1}^{k_n} \mathbb{V}\mathbf{ar}(Y_{n,i}) \to \sigma^2, \quad \text{as } n \to \infty.$$

Then $\sum_{i=1}^{k_n}(Y_{n,i} - \mathbb{E}[Y_{n,i}]) \xrightarrow{d} N(0,\sigma^2)$.

Theorem 22 (**Delta method**)**.** Suppose T_n is a estimator (a $\mathfrak{B}(\mathbb{R})$-measurable function of data independent of unknown parameter θ) of θ. Let $\phi : \mathbb{R}^k \to \mathbb{R}^m$ be a map defined on a subset of $\mathbb{R}^k$ and differentiable at θ, and T_n takes values inside the domain of ϕ. If $r_n(T_n - \theta) \xrightarrow{d} T$ as $n \to \infty$, then

$$r_n(\phi(T_n) - \phi(\theta)) \xrightarrow{d} \phi'_\theta \cdot T; \quad r_n(\phi(T_n) - \phi(\theta)) - \phi'_\theta(r_n(T_n - \theta)) \xrightarrow{P} 0.$$

Here ϕ'_θ is the Jacobian of ϕ evaluated at θ.

2 Basics of Statistical inference

2.1 Statistical Estimators

Definition 13 (**Statistics and estimator**). Given a random sample (an i.i.d collection) $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} f(x;\theta)$, we say a (measurable) function $T(\boldsymbol{X}) = T(X_1, \cdots, X_n)$ is an **estimator** of θ, if it does not depend on the unknown parameter θ. Such a $T(\boldsymbol{X})$ is also known as a **statistic**.

Definition 14 (**Unbiasedness**). An estimator $T(\boldsymbol{X})$ of θ is **unbiased**, if $\mathbb{E}_{X_1}[T(\boldsymbol{X})] = \theta$.

Definition 15 (**Consistency**). An estimator $T(\boldsymbol{X})$ of θ is **consistent**, if $T(\boldsymbol{X}) \xrightarrow{P} \theta$.

In the theory of point estimation, consider $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} f(x;\theta)$ where θ is one-dimensional. The following regularity conditions are introduced:

R1 The family $\left\{ f(x,\theta) : \theta \in \Theta \subset \mathbb{R} \right\}$ has a common support that does not depend on θ.

R2 $\dfrac{d}{d\theta} \log f(x,\theta)$ always exists.

R3 For any statistic $h(\boldsymbol{X}) \in L^1$,

$$\frac{d}{d\theta} \int_{\mathcal{S}} h(\boldsymbol{x}) f(\boldsymbol{x},\theta) dx = \int_{\mathcal{S}} h(\boldsymbol{x}) \frac{d}{d\theta} f(\boldsymbol{x},\theta) dx.$$

Theorem 23 (**Crémer-Rao Lower Bound**). Under regularly conditions [R1]-[R3] introduced above, given a random sample $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} f(x;\theta)$ with joint pdf (or pmf) denotes as $p_\theta(\boldsymbol{x})$. Suppose $T(\boldsymbol{X})$ is an unbiased estimator for $\tau(\theta)$, where $\tau : \mathbb{R} \to \mathbb{R}$ is a measurable function, then

$$\mathbb{V}\mathbf{ar}(T(\boldsymbol{X})) \geq \frac{[\tau'(\theta)]^2}{\mathbb{E}\left\{ \left(\dfrac{\partial}{\partial\theta} \log p_\theta(\boldsymbol{x}) \right)^2 \right\}}.$$

The quantity $\mathcal{I}_n = \mathbb{E}\left\{ \left(\dfrac{d}{d\theta} \log p_\theta(\boldsymbol{x}) \right)^2 \right\}$ is known as the **Fisher information**, and $\mathcal{I}_n = n\mathcal{I}_1$, where $\mathcal{I}_1$ is simply $\mathbb{E}[(\frac{\partial}{\partial\theta} f(x;\theta))^2]$.

Proposition 2 (**Bartlett Identities**). Under regularity conditions [R1]-[R3], we have

- (First Bartlett identity) $\mathbb{E}\left\{ \dfrac{d}{d\theta} \log p_\theta(\boldsymbol{x}) \right\} = 0, \forall\theta \in \Theta$;
- (Second Bartlett identity) $\mathbb{E}\left\{ \left(\dfrac{d}{d\theta} \log p_\theta(\boldsymbol{x}) \right)^2 \right\} = -\mathbb{E}\left\{ \dfrac{d^2}{d\theta^2} \log p_\theta(\boldsymbol{x}) \right\}$.

Definition 16 (**Sufficiency**). Suppose the random sample $X_1, \cdots, X_n$ has a joint pdf (pmf) $p_\theta(\boldsymbol{x})$, $\theta \in \Theta$. We say a statistic $T(\boldsymbol{X})$ is a **sufficient statistics** of θ, if the conditional distribution of $X_1, \cdots, X_n | T(\boldsymbol{X}) = t$ for any $\theta \in \Theta$, t does not depend on θ.

Example 3. Let $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} \text{Bern}(\theta)$, we claim that $T(\boldsymbol{X}) = \sum_{i=1}^{n} X_i$ is a sufficient statistic of θ, this is because

$$\begin{aligned} f(x_1, \cdots, x_n, \theta | T(\boldsymbol{X}) = t) &= \frac{p_\theta(X_1 = x_1, \cdots, X_n = x_n, \sum X_i = t)}{p_\theta(\sum X_i = t)} \\ &= \frac{\prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{n-x_i}}{\binom{n}{t}\theta^t(1-\theta)^{n-t}} \\ &= \begin{cases} \frac{1}{\binom{n}{t}} & \text{if } \sum_{i=1}^{n} X_i = t \\ 0 & \text{otherwise} \end{cases}, \end{aligned}$$

which does not depend on θ.

Note that a sufficient statistic can also be a vector. For example, later we will see if $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} N(\mu, \sigma^2)$, then a sufficient statistic of (μ, σ^2) is

$$\left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} (X_i - \overline{X}_n)^2 \right).$$

Proposition 3. Let $T(\boldsymbol{X})$ be a sufficient statistic of θ, g be a one-to-one transformation, then $g(T(\boldsymbol{X}))$ is a sufficient statistic of θ.

So in the previous Bernoulli example, $T(\boldsymbol{X}) = \frac{1}{n} \sum_{i=1}^{n} X_i$ is also a sufficient statistic of θ.

Theorem 24 (Neyman-Fisher Factorization Theorem). Consider a random sample $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} f(x; \theta)$ with a joint pdf (pmf) $p_\theta(\boldsymbol{x}) =$, a statistic $T(\boldsymbol{X})$ is sufficient of θ if and only if

$$p_\theta(x_1, \cdots, x_n) = g(T(\boldsymbol{x}), \theta) \cdot h(x_1, \cdots, x_n),$$

for all $\theta \in \Theta$, and function g depends on $T(\boldsymbol{x}), \theta$ while $h(\boldsymbol{x})$ does not depend on θ.

If we go back to the Bernoulli example: $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} \text{Bern}(\theta)$, then we see that

$$\begin{aligned} p_\theta(x_1, \cdots, x_n) &= \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i}, x_i \in \{0, 1\} \\ &= \theta^{\sum X_i} \cdot (1-\theta)^{n - \sum x_i}. \end{aligned}$$

We let the first term be $g(T(\boldsymbol{x}), \theta)$ and the second term be $h(\boldsymbol{x})$, and we apply Neyman-Fisher theorem, $T(\boldsymbol{X}) = \sum_{i=1}^{n} X_i$ is hence a sufficient statistic of θ.

Moreover, let us consider $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} \text{Unif}(0, \theta)$, $\theta > 0$, then its pdf is given by

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} & x \in (0, \theta) \\ 0 & \text{otherwise} \end{cases}.$$

Then we have

$$\begin{aligned} p_\theta(\boldsymbol{x}) &= \prod_{i=1}^{n} \frac{1}{\theta} \cdot \mathbb{1}_{0<x_i<\theta} \\ &= \left(\frac{1}{\theta}\right)^n \cdot \mathbb{1}_{0<X_{(n)}<\theta} \cdot \mathbb{1}_{0<X_{(1)}}, \end{aligned}$$

where $X_{(n)} = \max\{X_1, \cdots, X_n\}$, $X_{(1)} = \min\{X_1, \cdots, X_n\}$, and let the first two terms be $g(T(\boldsymbol{X}), \theta)$ and the last term be $h(\boldsymbol{x})$, then by Neyman-Fisher theorem, $T(\boldsymbol{X}) = X_{(n)}$ is a sufficient statistic of θ.

Definition 17 (**Exponential family**)**.** A random variable X is said to belong to **exponential family**, if the density of X can be written as

$$p(x;\theta) = h(x) \cdot \exp\Big(\eta(\theta)T(x) - A(\theta)\Big).$$

If $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} f(x;\theta)$ belong to exponential family, then the joint density is

$$p(\boldsymbol{x};\theta) = \prod_{i=1}^{n} h(x_i) \cdot \exp\Big(\eta(\theta)T(\boldsymbol{X}) - nA(\theta)\Big),$$

where $T(\boldsymbol{X}) = \sum_n T(x)$ is the sufficient statistic of θ.

Note that if θ is a vector, then so is $T(X)$, and we write $\eta(\theta)^\top T(X)$ (the dot product). In many cases, we work with the canonical form of exponential family:

$$p(x;\theta) = h(x) \exp\Big(\theta T(x) - A(\theta)\Big), \tag{2.1}$$

where we take $\eta(\theta)$ in the original form to be our parameter.

Proposition 4. Suppose $X \sim p(x;\theta)$ belongs to exponential family having the form (2.1), then

$$\mathbb{E}[T(X)] = \frac{\partial}{\partial\theta} A(\theta), \quad \mathbb{V}\mathbf{ar}(T(X)) = \frac{\partial^2}{\partial\theta^2} A(\theta).$$

Proof. Since $p(x;\theta)$ is a density function, so it integrates to 1, i.e,

$$\int h(x) \cdot \exp\Big(\theta T(x) - A(\theta)\Big) dx = 1.$$

After rearranging, we have

$$A(\theta) = \log \int h(x) \exp(\theta T(x)) dx,$$

where

$$\frac{dA(\theta)}{d\theta} = \frac{\frac{\partial}{\partial\theta} \int h(x) \exp(\theta T(x)) dx}{\int h(x) \exp(\theta T(x)) dx}. \tag{2.2}$$

Here we will assume the function h, T satisfies some regularity conditions so the Leibniz rule will apply, hence

$$\begin{aligned}\frac{\partial}{\partial\theta}\int h(x)\exp(\theta T(x))dx &= \int \frac{\partial}{\partial\theta}h(x)\exp(\theta T(x))dx \\ &= \int h(x)T(x)\exp(\theta T(x))dx.\end{aligned}$$

Then we may multiply $\exp(-A(\theta))$ on both the numerator and denominator of (2.2), and we have

$$\frac{dA(\theta)}{d\theta} = \frac{\int h(x)T(x)\exp(\theta T(x)-A(\theta))dx}{\int h(x)\exp(\theta T(x)-A(\theta))dx} \equiv \mathbb{E}[T(X)],$$

and $A''(\theta)$ is hence easy to compute and verify. □

> REMARK
>
> Note that $\mathbb{V}\mathbf{ar}(X) \geq 0$, so the above proposition also tells us $A(\theta)$ is convex.

Suppose g is convex, then given η_1, η_2, the Bregman Divergence is defined as

$$\rho(\eta_1, \eta_2) = g(\eta_2) - g(\eta_1) - \langle \nabla g(\eta_1), \eta_2 - \eta_1 \rangle.$$

In fact, if p, q belongs to the same exponential family, their Kullback–Leibler divergence satisfies

$$\mathrm{KL}(p||q) = \int p(x)\log\frac{p(x)}{q(x)}dx = A(\theta_2) - A(\theta_1) - \langle \nabla A(\theta_1), \theta_2 - \theta_1 \rangle.$$

Definition 18. A sufficient statistics of θ, $T(\boldsymbol{X})$ is a **minimum sufficient statistics** if for any other sufficient statistics of θ, $T^*(\boldsymbol{X})$, we have $T(\boldsymbol{X}) = \phi(T^*(\boldsymbol{X}))$ for some measurable function ϕ.

Theorem 25 (Lehmann-Scheffé). For a parametric family $p_\theta(\cdot)$, suppose a statistic $T(\boldsymbol{X})$ is such that $\forall x, y \in \mathcal{X}$, $T(\boldsymbol{x}) = T(\boldsymbol{y})$ iff $\dfrac{p_\theta(\boldsymbol{x})}{p_\theta(\boldsymbol{y})}$ does not depend on θ, then $T(\boldsymbol{X})$ is a minimum sufficient statistic. In fact any one-to-one function of a minimum sufficient statistics is also minimum sufficient.

Definition 19 (Completeness). Let X be a random variable with a pdf (pmf) belonging to a parametric family $\mathcal{F} : \{f_\theta : \theta \in \Theta\}$. This family is said to be complete, if for any measurable function g with $\mathbb{E}[g(x)]$ exists, we have

$$\mathbb{E}[g(x)] = 0, \forall\theta \in \Theta \implies \mathbb{P}(g(x) = 0) = 1 \text{ almost surely.}$$

A statistic $T(\boldsymbol{X})$ is complete, if its family of distributions is complete.

The following definition defines the "best" unbiased estimator in terms of minimum variance:

Definition 20. An estimator $T(\boldsymbol{X})$ of a real valued parameter $\theta \in \Theta \subset \mathbb{R}^d$ where $\tau(\theta) : \Theta \to \mathbb{R}$ is said to be a **UMVUE** (Uniformly Minimum Variance Unbiased Estimator) of $\tau(\theta)$ if:

- $\mathbb{E}_\theta[T(\boldsymbol{X})] = \tau(\theta), \forall \theta \in \Theta$;
- For any other unbiased estimator of $\tau(\theta)$, call it $T^*(\boldsymbol{X})$, where $\mathbb{E}_\theta[T^*(\boldsymbol{X})] = \tau(\theta), \forall \theta \in \Theta$, and
$$\mathbb{V}\mathbf{ar}_\theta\{T(\boldsymbol{X})\} \leq \mathbb{V}\mathbf{ar}_\theta\{T^*(\boldsymbol{X})\}, \forall \theta \in \Theta.$$

Theorem 26 (Rao-Blackwell). Let $U(\boldsymbol{X})$ be an unbiased estimator for $\tau(\theta)$, let $T(\boldsymbol{X})$ be a complete sufficient statistic for the parametric family, then
$$\mathbb{E}[U(\boldsymbol{X})|T(\boldsymbol{X}) = t], \forall t \in \mathcal{S}_T$$
is the UMVUE for $\tau(\theta)$. If the sufficient statistic is not complete, then conditioning on it will only get a better estimator, and it is not necessarily the UMVUE. If $\widehat{\theta}$ is simply an estimator (not necessarily unbiased), then $\text{MSE}(\mathbb{E}[\widehat{\theta}|T(\boldsymbol{X})]) \leq \text{MSE}(\widehat{\theta})$.

Theorem 27 (Lehmann-Scheffé). Let $T(\boldsymbol{X})$ be a complete sufficient statistic, also let $U(\boldsymbol{X}) = h(T(\boldsymbol{X}))$ for a measurable function h, be an unbiased estimator of $\tau(\theta)$ such that $\mathbb{E}[U^2(\boldsymbol{X})] < \infty$. Then $U(\boldsymbol{X})$ is the unique UMVUE.

2.2 Empirical cumulative distribution functions (ECDF)

Below, denote $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} F$, where F is the CDF of X, where
$$F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} dF(t),$$
i.e, integrating with respect to measure F (could be continuous or discrete) on $(\mathbb{R}, \mathfrak{B}(\mathbb{R}))$.

Definition 21. The **empirical cumulative distribution function (ECDF)** of X is given by
$$\widehat{F}(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{X_i \leq x}.$$

Some properties of the ECDF follow natually from the definition.

Proposition 5. Let $\widehat{F}(x)$ be the ECDF of X, then

- **Unbiasedness**: $\mathbb{E}[\widehat{F}(x)] = F(x)$;
- $\mathbb{V}\mathbf{ar}(\widehat{F}(x)) = \frac{1}{n}F(x)(1 - F(x))$;
- Let $x, y \in \mathbb{R}$, then $\mathbb{C}\mathbf{ov}(\widehat{F}(x), \widehat{F}(y)) = \frac{1}{n}(F(\min\{x, y\}) - F(x)F(y))$.

Proof. The first two is trivial since we notice that $\mathbb{1}_{X_i \leq x} \sim \text{Bern}(p)$ where $p = \mathbb{P}(X_i \leq x) = F(x)$, and the result follows from the fact that $\widehat{F}(x)$ is the sample mean of i.i.d $\text{Bern}(p)$.

For the covariance part, first note that
$$\mathbb{C}\mathbf{ov}(\widehat{F}(x), \widehat{F}(y)) = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} \mathbb{C}\mathbf{ov}(\mathbb{1}_{X_i \leq x}, \mathbb{1}_{X_j \leq y}).$$

- If $i = j$, then

$$\begin{aligned}\mathbf{Cov}(\mathbb{1}_{X_i \le x}, \mathbb{1}_{X_j \le y}) &= \mathbb{E}[\mathbb{1}_{X_i \le x}\mathbb{1}_{X_j \le y}] - \mathbb{E}[\mathbb{1}_{X_i \le x}]\mathbb{E}[\mathbb{1}_{X_j \le y}] \\ &= \mathbb{E}[\mathbb{1}_{X_i \le x}\mathbb{1}_{X_i \le y}] - F(x)F(y) \\ &= \mathbb{E}[\mathbb{1}_{X_i \le \min\{x,y\}}] - F(x)F(y) \\ &= F(\{x, y\}) - F(x)F(y).\end{aligned}$$

- If $i \neq j$, by i.i.d assumption, we simply have $\mathbf{Cov}(\mathbb{1}_{X_i \le x}, \mathbb{1}_{X_j \le y}) = 0$.

Thus combining the results above yields

$$\mathbf{Cov}(\widehat{F}(x), \widehat{F}(y)) = \frac{1}{n}(F(\min\{x, y\}) - F(x)F(y)).$$

□

Also since $\mathbb{V}\mathbf{ar}(\widehat{F}(x)) \to 0$ as $n \to \infty$, combine with unbiasedness, we know that

$$\widehat{F}(x) \xrightarrow{P} F(x), \quad \text{by Chebyshev inequality.}$$

Also by using central limit theorem, we know that

$$\sqrt{n}(\widehat{F}(x) - F(x)) \xrightarrow{d} N(0, F(x)(1 - F(x)).$$

Using chebyshev inequality, we are able to obtain convergence in probability of the ECDF. In fact stronger convergence could be obtained.

Theorem 28 (**Glivenko-Cantelli theorem**)**.** Let $\widehat{F}(x)$ be the ECDF of X, then

$$\sup_{x \in \mathbb{R}} |\widehat{F}(x) - F(x)| \xrightarrow{\text{a.s}} 0.$$

We refer to the proof given by Klenke [3] (Theorem 5.23)

Proof. Fix $x \in \mathbb{R}$ and for all $n \in \mathbb{N}$, define

$$Y_n(x) = \mathbb{1}_{(-\infty, x]}(X_n), \qquad Z_n(x) = \mathbb{1}_{(-\infty, x)}(X_n).$$

Additionally, define $F(x-) = \lim_{y \uparrow x} F(y)$, and similarly for F_n. Then by i.i.d assumption, each of $\{Y_n\}, \{Z_n\}$ is independent. Note that $\mathbb{E}[Y_n(x)] = \mathbb{P}(X_n \le x) = F(x)$, $\mathbb{E}[Z_n(x)] = \mathbb{P}(X_n < x) = F(x-)$. By the strong law of large numbers, we thus have

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} Y_i(x) \xrightarrow[n \to \infty]{\text{a.s.}} F(x), \quad F_n(x-) = \frac{1}{n}\sum_{i=1}^{n} Z_i(x) \xrightarrow[n \to \infty]{\text{a.s.}} F(x-).$$

Formally, define $F(-\infty) = 0, F(\infty) = 1$, fix some ($N \in \mathbb{N}$ and define

$$x_j := \inf\left\{x \in \overline{\mathbb{R}} : F(x) \ge \frac{j}{N}\right\}, \quad j = 0, \ldots, N,$$

and

$$R_n := \max_{j=1,\dots,N-1} \left(\left| F_n(x_j) - F(x_j) \right| + \left| F_n(x_j-) - F(x_j-) \right| \right).$$

As the consequence of S.L.L.N, $R_n \xrightarrow[n\to\infty]{\text{a.s.}} 0$. For $x \in (x_{j-1}, x_j)$, we have, by the definition of x_j,

$$F_n(x) \le F_n(x_j-) \le F(x_j-) + R_n \le F(x) + R_n + \frac{1}{N},$$

and

$$F_n(x) \ge F_n(x_{j-1}) \ge F(x_{j-1}) - R_n \ge F(x) - R_n - \frac{1}{N}.$$

Hence

$$\limsup_{n\to\infty} \sup_{x\in\mathbb{R}} |F_n(x) - F(x)| \le \frac{1}{N} + \limsup_{n\to\infty} R_n = \frac{1}{N}$$

almost surely. Since $N \in \mathbb{N}$ was arbitrary, letting $N \to \infty$ gives the desired result. □

Theorem 29 (**Dvoretzky-Kiefer-Wolfowitz**)**.** Let $\widehat{F}$ be the ECDF of X, then $\forall \epsilon > 0$,

$$\mathbb{P}\left(\sup_{x\in\mathbb{R}} |\widehat{F}(x) - F(x)| \ge \epsilon \right) \le 2e^{-2n\epsilon^2}.$$

Note that when the sup condition is dropped, the above result is simply an application of Hoeffding's inequality.

We often encounter estimands that is a function of the CDF, $T(F)$. To estimate $T(F)$, a naïve way is to replace F by its empirical estimate $\widehat{F}$, which is called the plug in estimator of $\theta = T(F)$, and denote as $\widehat{\theta} = T(\widehat{F})$.

Definition 22. A **linear functional** of F is a functional T such that

$$T(F) = \int r(x) dF(x),$$

for some function r.

If T is linear, then the plug in estimator of $T(F)$ is simply

$$T(\widehat{F}) = \int r(x) d\widehat{F}(x) = \frac{1}{n} \sum_{i=1}^{n} r(X_i),$$

and in the sense of ECDF, the function r is simply the indicator.

Definition 23. The **Gâteaux derivative** of a functional T at F in the direction of G is given by

$$L_F(G) = \lim_{\epsilon\to 0} \frac{T((1-\epsilon)F + \epsilon G) - T(F)}{\epsilon}.$$

Suppose $G(t) = \mathbb{1}_{x \le t}$, then for notation simplicity we denote $G = \delta_x$ (G has unit mass at x). Then $L_F(x) := L_F(\delta_x)$ is called the **influence function**. It represents the change in $T(F)$ when an infinitesimal amount ϵ of mass is subtracted from F, and replaced with a point mass at x. We have

$$L_F(x) = \lim_{\epsilon \to 0} \frac{T((1-\epsilon)F + \epsilon \delta_x) - T(F)}{\epsilon},$$

where we may replace F by $\widehat{F}$ to get the estimate $\widehat{L}_F(x)$. Usually, we drop the subscript and denote $L(x)$ as the influence function when the context of F is clear.

Proposition 6. Let T be a linear functional (i.e, $T(aF + bG) = aT(F) + bT(G)$). Then for any G,

- $L_F(G) = T(G) - T(F)$;
- $L_F(x) = r(x) - T(F)$;
- $T(G) = T(F) + \int L_F(x) dG(x)$;
- $\int L_F(x) dF(x) = 0$.

Proof. .

- By definition,

$$\begin{aligned} L_F(G) &= \lim_{\epsilon \to 0} \frac{T((1-\epsilon)F + \epsilon G) - T(F)}{\epsilon} \\ &= \lim_{\epsilon \to 0} \frac{T(F) - \epsilon T(F) + \epsilon T(G) - T(F)}{\epsilon} \\ &\equiv T(G) - T(F). \end{aligned}$$

- By definition,

$$L_F(x) = \int r(t) d\delta_x(t) dt - T(F) \equiv r(x) - T(F).$$

- Use the fact that $L_F(x) = r(x) - T(F)$ and the integral definition yields the result immediately.
- Same as the previous one.

□

If we let $G := \widehat{F}$, the ECDF, then according to Proposition 6,

$$T(\widehat{F}) - T(F) = L_F(\widehat{F}) = \frac{1}{n} \sum_{i=1}^{n} L_F(X_i) = \int L_F(x) dF(x) \equiv 0.$$

If

$$\tau^2 := \int L_F^2(x) dF(x) = \int (r(x) - T(F))^2 dF(x) < \infty,$$

then by central limit theorem, we have

$$\sqrt{n}(T(\widehat{F}) - T(F)) \xrightarrow{d} N(0, \tau^2),$$

where τ^2 can be estimated by

$$\widehat{\tau^2} = \frac{1}{n}\sum_{i=1}^{n} \widehat{L_F^2}(X_i) = \frac{1}{n}\sum_{i=1}^{n}(r(X_i) - T(\widehat{F}))^2 \xrightarrow{P} \tau^2,$$

and using Slutsky's theorem,

$$\sqrt{n} \cdot \frac{T(\widehat{F}) - T(F)}{\widehat{\tau}} \xrightarrow{d} N(0,1).$$

Theorem 30. If T is Hadamard differentiable[2] with respect to metric $d(F,G) = \sup_x |F(x) - G(x)|$, then

$$\sqrt{n}(T(\widehat{F}_n) - T(F)) \xrightarrow{d} N(0, \tau^2),$$

where $\tau^2 = \int L_F(x)^2 dF(x)$. The approximation

$$\frac{T(\widehat{F}_n) - T(F)}{\widehat{\text{se}}} \xrightarrow{d} N(0,1), \quad \widehat{\text{se}} = \frac{\widehat{\tau}}{n} = \frac{\left(\frac{1}{n}\sum_{i=1}^{n} \widehat{L}^2(X_i)\right)^{1/2}}{\sqrt{n}}$$

is called the **non-parametric delta method**.

Example 4 (Population mean). Suppose $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} F$ where $\theta = T(F) = \int x dF(x)$ is the parameter of interest (the population mean). Then obviously $T(F)$ is a linear functional by definition, also we know an unbiased estimator of θ is the ECDF $\widehat{F}_n$. Hence the plug in estimator of θ is

$$\widehat{\theta} = T(\widehat{F}_n) = \int x d\widehat{F}_n := \overline{X}_n.$$

Further, the influence function is given by

$$L(x) = \lim_{\epsilon \to 0} \frac{T((1-\epsilon)F + \epsilon\delta_x) - T(F)}{\epsilon} = \lim_{\epsilon \to 0} \frac{(1-\epsilon)\theta + \epsilon x}{\epsilon} = x - \theta,$$

hence $\widehat{L}(x) = x - \overline{X}_n$, and

$$\widehat{\tau^2} = \int \widehat{L}(x) d\widehat{F}_n = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2,$$

which means the estimated standard deviation is given by $\widehat{\text{se}} = \frac{\widehat{\tau}}{n}$, whereas we may then construct confidence intervals for $\theta = T(F)$ given significance level $1 - \alpha$:

$$\left(T(\widehat{F}) - z_{\alpha/2} \cdot \widehat{\text{se}} \ , \ T(\widehat{F}) + z_{\alpha/2} \cdot \widehat{\text{se}}\right).$$

[2] Let $T : D \subseteq E \to B$, be a map between Banach spaces. The functional T is Hadamard differentiable at $F \in D$ if there exists a continuous linear map $DT_F : E \to B$ such that $\frac{T(F + t_n h_n) - T(F)}{t_n} \to DT_F(h)$ whenever $t_n \to 0$, $t_n \neq 0$, $h_n \to h$, and $F + t_n h_n \in D$.

When the statistical functional is multivariate $T(F) = a(T_1(F), T_2(F), \cdots, T_m(F))$, the influence function is obtained via chain rule

$$L(x) = \sum_{i=1}^{m} \frac{\partial a}{\partial t_i} L_i(x),$$

where

$$L_i(x) = \lim_{\epsilon \to 0} \frac{T_i((1-\epsilon)F + \epsilon\delta_x) - T_i(F)}{\epsilon}.$$

Example 5 (Non-linear functional). If we consider the parameter of interest is the quantile function $\theta = T(F) = F^{-1}(p)$, then $T(F)$ is not a linear functional, hence we cannot use Proposition 6 to get the influence function. Instead, let

$$F_\epsilon(y) = (1-\epsilon)F(y) + \epsilon\delta_x(y),$$

where $\delta_x(y) = \mathbb{1}_{y \geq x}$. Since $F_\epsilon(T(F_\epsilon)) \equiv p$, hence $\frac{\partial}{\partial \epsilon} F_\epsilon(T(F_\epsilon)) = 0$. By chain rule,

$$\begin{aligned}\frac{\partial}{\partial \epsilon} F_\epsilon(T(F_\epsilon)) &= \frac{\partial}{\partial \epsilon}(1-\epsilon)F(T(F_\epsilon)) + \epsilon\delta_x(T(F_\epsilon)) \\ &= -F(T(F_\epsilon)) + (1-\epsilon) \cdot f(T(F_\epsilon)) \cdot \frac{\partial}{\partial \epsilon} F(T(F_\epsilon)) + \delta_x(T(F_\epsilon)) + \epsilon \frac{\partial}{\partial \epsilon}\delta_x(T(F_\epsilon)).\end{aligned}$$

Note that $F_\epsilon(y)\Big|_{\epsilon=0} \equiv F(y)$, so

$$\frac{\partial}{\partial \epsilon} F_\epsilon(T(F_\epsilon))\Big|_{\epsilon=0} = -p + f(T(F)) \cdot \frac{\partial}{\partial \epsilon} T(F)\Big|_{\epsilon=0} + \delta_x(T(F_\epsilon)) + \epsilon \cdot \frac{\partial}{\partial \epsilon}\delta_x(T(F_\epsilon))\Big|_{\epsilon=0} = 0.$$

Also note that $T(F) = \theta = F^{-1}(p)$, $\frac{\partial}{\partial \epsilon} T(F)\Big|_{\epsilon=0} = L_F(x)$, we hence have

$$0 = -p + f(\theta) \cdot L_F(x) + \delta_x(\theta),$$

therefore

$$L_F(x) = \begin{cases} \frac{p-1}{f(\theta)} & \text{if } x \leq \theta \\ \frac{p}{f(\theta)} & \text{if } x > \theta \end{cases}.$$

2.3 Bootstrap

Suppose we have a statistic $T_n = g(X_1, \cdots, X_n)$ where $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} F$ and F is unknown. If $T_n = \bar{X}_n$, then the variance of T_n under F can be easily expressed. However, when T_n is complex, we might not be able to express the variance in closed form. If we assume g is some integrable measurable function ($\mathbb{E}|g(X)| < \infty$), then the law of large numbers tells us

$$\frac{1}{n}\sum_{i=1}^{n} g(X_i) \xrightarrow{p} \int g(x) dF(x) := \mathbb{E}[g(X)],$$

which is the idea behind Monte-Carlo integration. The bootstrap uses a similar idea, where the variance is obtained by repeated sampling and averaging.

Given a random sample $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} F$ with a statistic T_n our goal is to estimate $\mathbb{V}\mathbf{ar}_F(T_n)$. To do so one might want to repeatedly draw i.i.d samples from F and estimate the variance by averaging. However F is unknown. The idea is to replace F by ECDF $\widehat{F}_n$, and if $\widehat{F}_n$ is "close to" F then so should $\mathbb{V}\mathbf{ar}_{\widehat{F}_n}(T_n)$ to $\mathbb{V}\mathbf{ar}_F(T_n)$. Note that under the discrete distribution $\widehat{F}_n$, each observation has mass $\frac{1}{n}$. So drawing i.i.d samples from $\widehat{F}_n$ is the same as drawing samples uniformly from $X_1, \cdots, X_n$ with replacement, and this is the idea of the **non-parametric bootstrap**. We denote the rth i.i.d bootstrap sample from $\widehat{F}_n$ as $X^*_{r,1}, X^*_{r,2}, \cdots, X^*_{r,n}$. Then we may use $T^*_{n,r} = g(X^*_1, \cdots, X^*_n)$ to denote the bootstrap statistic. If the total size of Bootstrap sample is B, then the variance of T_n can be estimated by

$$(\mathbb{V}\mathbf{ar}_F(T_n))_{\text{boot}} = \frac{1}{B}\sum_{b=1}^{B}(T^*_{n,b} - \overline{T}^*_n)^2, \quad \text{where } \overline{T}^*_n = \frac{1}{B}T^*_{n,r}. \tag{2.3}$$

Proposition 7. Let $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} F$, $X^*_1, \cdots, X^*_n$ be a bootstrap sample from $\widehat{F}_n$. Then

- $\mathbb{E}[\bar{X}^*_n | X_1, \cdots, X_n] = \overline{X}_n$;
- $\mathbb{V}\mathbf{ar}(\bar{X}^*_n | X_1, \cdots, X_n) = \dfrac{1}{n^2}\left(\sum_{j=1}^{n} X_j^2 - n\overline{X}_n^2\right) \equiv \dfrac{n-1}{n^2}S_n^2$.

Proof. It is easy to see that

$$\begin{aligned}
\mathbb{E}[\bar{X}^*_n | X_1, \cdots, X_n] &= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X^*_i | X_1, \cdots, X_n] \\
&= \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} X_j \cdot \mathbb{P}(X^*_i = x_j | X_1, \cdots, X_n) \\
&= \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} X_j \cdot \frac{1}{n} \\
&\equiv \overline{X}_n.
\end{aligned}$$

For variance, we have

$$\begin{aligned}
\mathbb{V}\mathbf{ar}(\bar{X}^*_n | X_1, \cdots, X_n) &= \mathbb{E}[\bar{X}^{*2}_n | X_1, \cdots, X_n] - \Big(\mathbb{E}[\bar{X}_n | X_1, \cdots, X_n]\Big)^2 \\
&= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\mathbb{E}[X^*_i X^*_j | X_1, \cdots, X_n] - (\overline{X}_n)^2.
\end{aligned}$$

If $i = j$ (a total of n cases), then

$$\mathbb{E}[X^*_i X^*_j | X_1, \cdots, X_n] = \sum_{i=1}^{n} X_i^2 \cdot \mathbb{P}(X^*_i = X_j | X_1, \cdots, X_n) = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

On the other hand, let $i \neq j$ (a total of $n(n-1)$ cases), then by independence, we have

$$\mathbb{E}[X_i^* X_j^*] = \mathbb{E}[X_i^* | X_1, \cdots, X_n] \cdot \mathbb{E}[X_j^* | X_1, \cdots, X_n] = \overline{X}_n^2.$$

Thus combining them yields

$$\mathbb{V}\mathbf{ar}(\bar{X}_n^* | X_1, \cdots, X_n) = \frac{1}{n^2}\left(\sum_{i=1}^n X_i^2 + n(n-1)\overline{X}_n^2\right) - \overline{X}_n^2.$$

□

In fact, by using the law of iterated expectation and variance, we have

$$\mathbb{E}[\bar{X}_n^*] = \mathbb{E}[\mathbb{E}[\bar{X}_n^* | X_1, \cdots, X_n]] = \mathbb{E}[X_1],$$

and

$$\mathbb{V}\mathbf{ar}(\bar{X}_n^*) = \mathbb{E}[\mathbb{V}\mathbf{ar}(\bar{X}_n^* | X_1, \cdots, X_n)] + \mathbb{V}\mathbf{ar}(\mathbb{E}[\bar{X}_n^* | X_1, \cdots, X_n]) = \frac{\mathbb{V}\mathbf{ar}(X_1)}{n} + \frac{(n-1)\mathbb{V}\mathbf{ar}(X_1)}{n^2}.$$

Based on simulation 2.3, We may then construct confidence intervals for the bootstrap variance estimate. The formal definition of a confidence interval is as follow.

Definition 24. Let $L(\boldsymbol{X}), U(\boldsymbol{X})$ be two statistics such that $L(\boldsymbol{x}) \leq U(\boldsymbol{x}), \forall x \in \mathcal{X}$. A random interval $(L(\boldsymbol{X}), U(\boldsymbol{X}))$ is called a **confidence interval** with significance level $1-\alpha, \alpha \in (0,1)$ if $P(L(\boldsymbol{X}) \leq \theta \leq U(\boldsymbol{X})) = 1-\alpha$.

One way to construct confidence interval is by using pivotal quantity.

Definition 25 (**Pivotal interval**)**.** A random function $Q(\boldsymbol{x}, \theta)$ is called a **pivot quantity** (PQ) iff its distribution does not depend on the parameter θ.

Note that the function Q might include θ, but its overall distribution $Q(\boldsymbol{x}, \theta)$ is free of any parameters. For example if $X \sim N(0, \theta^2)$, then $Q(x, \theta) = \frac{1}{\theta} X \sim N(0,1)$, which is free of any parameter and it is a known distribution, hence $Q(x, \theta)$ is a pivot quantity. Once we have a PQ, and the confidence level $1-\alpha$ is given, we may find constants c_1, c_2 such that

$$\mathbb{P}(c_1 \leq Q(\boldsymbol{x}, \theta) \leq c_2) = 1 - \alpha,$$

then we can solve the equation above in terms of θ to get the confidence interval

$$\mathbb{P}(L(\boldsymbol{X}) \leq \theta \leq U(\boldsymbol{X})) = 1 - \alpha.$$

Suppose $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} F$, and our parameter of interest is a functional of F, say $\theta = T(F)$. Let $\widehat{\theta}_n = T(\widehat{F}_n)$ where $\widehat{F}_n$ is the ECDF. The pivot Q_n is defined as

$$Q_n := \widehat{\theta}_n - \theta.$$

Let $H(x) = \mathbb{P}(Q_n \leq x)$, the CDF of the pivot, then the $1-\alpha$ confidence interval can be obtained by

$$\theta \in \left(\widehat{\theta}_n - H^{-1}(1-\alpha/2) \ , \ \widehat{\theta}_n - H^{-1}(\alpha/2)\right).$$

In practice, we don't usually know H, instead we will estimate H via bootstrapping. Denote $X^*_{r,1}, \cdots, X^*_{r,n} \overset{\text{i.i.d}}{\sim} \widehat{F}_n$ as the rth bootstrap sample, then let $Q^*_{n,r} = \widehat{\theta}^*_{n,r} - \widehat{\theta}_n$. The estimate of H is given by

$$\widehat{H}(x) = \frac{1}{B} \sum_{b=1}^{B} \mathbb{1}_{Q^*_{n,b} \leq x},$$

where B is the total number of bootstrap samples. Denote q^*_p as the pth quantile of $(Q^*_{n,1}, \cdots, Q^*_{n,B})$, θ^*_p as the pth quantile of $(\widehat{\theta}^*_{n,1}, \cdots, \widehat{\theta}^*_{n,B})$, then $q^*_p = \theta^*_p - \widehat{\theta}_n$, and the estimated confidence interval is hence

$$\theta \in \left(\widehat{\theta}_n - q^*_{1-\alpha/2} \ , \ \widehat{\theta}_n - q^*_{\alpha/2}\right) = \left(2\widehat{\theta}_n - \theta^*_{1-\alpha/2} \ , \ 2\widehat{\theta}_n - \theta^*_{\alpha/2}\right).$$

3 Point Estimation

3.1 Method of Moments

In methods of moments, we simply match the kth population moment with the kth sample moment. Let $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} f(\theta, x)$ where $\theta \in \Theta \subset \mathbb{R}^d$, so our parameter of interest is $\boldsymbol{\theta} = (\theta_1, \cdots, \theta_d)^\top$. With the assumption that $\mathbb{E}|X_i|^d$ exists, we denote the jth population moments as

$$\mu_j(\boldsymbol{\theta}) = \mathbb{E}X_i^j$$

and the jth sample moments as

$$m_j(\boldsymbol{X}) = \frac{1}{n}\sum_{i=1}^n X_i^j.$$

Definition 26. In methods of moment estimator, we will match each population moment and sample moment and solve the corresponding systems of equations. That is, we set

$$m_j(\boldsymbol{X}) = \frac{1}{n}\sum_{i=1}^n X_i^j \overset{\text{set}}{=} \mu_j(\boldsymbol{\theta}) = \mathbb{E}X_i^j$$

where $j = 1, \cdots, d$ and solve the corresponding system of equations.

We need to note that in general $m_j(\boldsymbol{X}) \neq \mu_j(\boldsymbol{\theta})$, it is only an estimate. Hence by solving the system of equations, we will have the estimate of θ_j as a function of $m_1, \cdots, m_d$:

$$\widehat{\theta}_j = g_j(m_1, \cdots, m_d)$$

Example 6. Find the methods of moments estimator of a normal random sample $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} N(\mu, \sigma^2)$.

Here we have two parameters so we have $\boldsymbol{\theta} = (\mu, \sigma^2)^\top$ and we match each sample moment with population moment up to $d = 2$:

$$m_1(\boldsymbol{X}) = \frac{1}{n}\sum_{i=1}^n X_i = \mu_1(\boldsymbol{\theta}) = \mathbb{E}X_i = \mu;$$

$$m_2(\boldsymbol{X}) = \frac{1}{n}\sum_{i=1}^n X_i^2 = \mu_2(\boldsymbol{\theta}) = \mathbb{E}X_i^2 = \sigma^2 + \mu^2.$$

So if we solve the above system of equations, we will get

$$\begin{cases} \widehat{\mu} & = \overline{X}_n \\ \\ \widehat{\sigma}^2 & = \dfrac{1}{n}\displaystyle\sum_{i=1}^n \left(X_i - \overline{X}_n\right)^2 \end{cases}$$

as the methods of moments estimator. Notice here the estimate of σ^2 is biased, and the unbiased estimate of σ^2 is in fact $\dfrac{1}{n-1}\displaystyle\sum_{i=1}^n (X_i - \overline{X}_n)^2$. But asymptotically, using weak law of large

numbers and continuous mapping theorem, our methods of moments estimate will converge to σ^2 in probability.

Methods of moments estimator may not always exist, since we need to assume the existence of $\mathbb{E}|X_i|^d$.

3.2 Maximum likelihood estimator (MLE)

Definition 27. Consider the random sample $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} f(\theta, x)$ with joint pdf (pmf) $p(\boldsymbol{x}; \theta)$, the likelihood function $L(\theta)$ is defined as $L(\theta) = p(\boldsymbol{x}; \theta)$, and the log-likelihood function is denote as $\ell(\theta) = \log p(\boldsymbol{x}; \theta)$.

The likelihood principle says that, in light of the data $\boldsymbol{x}$, the likelihood function contains all the information in the data about θ, two likelihoods function contain the same information about θ if they are proportional to each other.

Definition 28. Given $\boldsymbol{x} = (x_1, \cdots, x_n)$, $\widehat{\theta}_n$ which is a function of $\boldsymbol{x}$ is called the maximum likelihood estimate (MLE) of θ if it maximizes $L(\theta)$ or $\ell(\theta)$, i.e

$$\widehat{\theta}_n = \widehat{\theta}(x_1, \cdots, x_n) = \arg\max_{\theta \in \Theta} L(\theta) = \arg\max_{\theta \in \Theta} \ell(\theta).$$

The term $\widehat{\theta}(X_1, \cdots, X_n)$ is called the **maximum likelihood estimator** of θ.

The interpretation of MLE is that we want to select a value of $\theta \in \Theta$ that maximizes the probability of observing what we observed as $\boldsymbol{x}$. Here are some remarks:

(i) The MLE of θ may not exist when Θ is open. It always exist over the closure of Θ;

(ii) If Θ is finite, then $\Theta = \bar{\Theta}$ and the MLE always exists and can be computed by comparing the values of the likelihood (log-likelihood) function;

(iii) If $L(\theta)$ or $\ell(\theta)$ is differentiable on Θ, then possible candidates of MLE are values of $\theta \in \Theta$ such that the first derivative of the likelihood (log-likelihood) function is zero, we may do a second derivative test to further verify if the function is twice differentiable. If the function is not differentiable, then extrema may occur at non-differentiable or discontinuity points, hence it is important to analyze the entire likelihood function to find its maximum.

Example 7. Find the MLE of the random sample $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} \text{Pois}(\lambda), \lambda > 0$.

First the joint pdf is given by

$$p_\lambda(\boldsymbol{x}) = \prod_{i=1}^{n} e^{-\lambda} \frac{\lambda^{x_i}}{x_i!}$$

hence

$$\ell_n(\theta) = -\sum_{i=1}^{n} \log(x_i!) - n\theta + n\bar{x}_n \log\theta, \quad \frac{d}{d\theta}\ell_n(\theta) = -n + n\bar{x}_n \frac{1}{\theta}$$

set $\frac{d}{d\theta}\ell_n(\theta) = 0$, we have $\widehat{\theta}_n = \bar{x}_n$. By second derivative test,

$$\frac{d^2}{d\theta^2}\ell_n(\theta)\bigg|_{\theta=\bar{x}_n} = -\frac{n\bar{x}_n}{\theta^2} < 0$$

which means $\widehat{\theta}_n = \bar{x}_n$ is the maximum likelihood estimate of the sample, then the maximum likelihood estimator is just $\widehat{\theta}(X_1, \cdots, X_n) = \bar{X}_n$.

The next example demonstrates the case when the likelihood function is not differentiable on $\mathbb{R}$:

Example 8. Find the MLE of $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} \text{Unif}(0, \theta), \theta > 0$.

In this case, the joint likelihood is given by

$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} \cdot \mathbb{1}\{(x_i \in [0, \theta]\} := \frac{1}{\theta} \cdot \mathbb{1}\{x_{(n)} \leq \theta\}, \quad x_i \geq 0,$$

where $x_{(n)} := \max_n\{x_1, \cdots, x_n\}$. As a function of $\theta > 0$, we note that

$$\frac{1}{\theta} \cdot \mathbb{1}_{x_{(n)} \leq \theta} = \begin{cases} 0 & \text{if } \theta < x_{(n)} \\ \frac{1}{\theta} & \text{if } \theta \geq x_{(n)} \end{cases}.$$

So $L(\theta)$ is constant 0 on $(0, x_{(n)})$ and is $\frac{1}{\theta}$ on $[\theta, \infty)$. Hence the maximum of $L(\theta)$ is obtained at $\widehat{\theta} = X_{(n)}$, which is the MLE of θ.

We next study some properties of the MLE.

Theorem 31 (**Consistency**)**.** If $\widehat{\theta}_n$ is the MLE of θ, then $\widehat{\theta}_n \xrightarrow{P} \theta$.

Theorem 32 (**Invariance**)**.** If $\widehat{\theta}_n$ is the MLE of θ, then for any function $\tau(\theta)$, $\tau(\widehat{\theta}_n)$ is the MLE for $\tau(\theta)$.

Theorem 33 (**Asymptotic Normality**)**.** Under regularity conditions, if $\widehat{\theta}_n$ is the MLE of θ, then

$$\sqrt{n}(\widehat{\theta}_n - \theta) \xrightarrow{d} N(0, \mathcal{I}_1^{-1}(\theta)),$$

where $\mathcal{I}_1(\theta)$ is the Fisher information of X_1. If g is a measurable function such that $g'(\theta) \neq 0$ and exists, then

$$\sqrt{n}(g(\widehat{\theta}_n - g(\theta)) \xrightarrow{d} N(0, [g'(\theta)]^2 \cdot \mathcal{I}_1^{-1}(\theta)).$$

Note that by law of large numbers and continuous mapping theorem, we have that

$$[g'(\widehat{\theta}_n)]^2 \xrightarrow{P} g'(\theta)^2; \quad \mathcal{I}_1^{-1}(\widehat{\theta}_n) \xrightarrow{P} \mathcal{I}_1^{-1}(\theta).$$

Hence, by Slutsky's theorem, we also have

$$\sqrt{n}(g(\widehat{\theta}_n - g(\theta)) \xrightarrow{d} N(0, [g'(\theta)]^2 \cdot \mathcal{I}_1^{-1}(\theta)),$$

which is helpful in constructing confidence intervals for θ:

Example 9. Suppose we have a random sample $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} \text{Pois}(\lambda)$ and λ is the unknown parameter. Using the asymptotic normality of MLE, construct a $100(1-\alpha)\%$ approximate two-sided confidence interval for λ, λ^2, and $e^{-\lambda}$.

Here, we can easily see that the MLE of a Poisson random sample is just the sample mean, $\widehat{\theta}_n = \overline{X}_n$. Now we find the Fisher information. The log-pdf of one sample is

$$\log f(X = k, \lambda) = \log e^{-\lambda} \frac{\lambda^k}{k!} = -\lambda + k \log \lambda - \log k!.$$

The second partial derivative is

$$\frac{\partial^2 \log f(X = k, \lambda)}{\partial \lambda^2} = -k \frac{1}{\lambda^2},$$

and hence we have

$$\mathcal{I}_1(\lambda) = -\mathbb{E}\left[-X \frac{1}{\lambda^2}\right] = \frac{1}{\lambda}.$$

So the MLE theory says that

$$\sqrt{n}\left(\overline{X}_n - \lambda\right) \xrightarrow{d} N\left(0, \mathcal{I}_1^{-1}(\widehat{\theta}_n)\right).$$

Given that $\mathcal{I}_1^{-1}(\widehat{\theta}) = \overline{X}_n$, hence the $100(1-\alpha)\%$ C.I is

$$\left(\overline{X}_n - z_{\alpha/2} \cdot \sqrt{\frac{1}{n} \cdot \overline{X}_n} \ , \ \overline{X}_n + z_{\alpha/2} \cdot \sqrt{\frac{1}{n} \cdot \overline{X}_n}\right).$$

For λ^2 and $e^{-\lambda}$, we will then apply delta-method to find the C.I, we define $g(\lambda) = \lambda^2$, and $h(\lambda) = e^{-\lambda}$, hence it is easy to see that $\widehat{\lambda^2}_{MLE} = \overline{X}_n^2$ and $\widehat{e^{-\lambda}}_{MLE} = e^{-\overline{X}_n}$. Then delta-method says that

$$\sqrt{n}(\overline{X}_n^2 - \lambda^2) \xrightarrow{d} N\left(0, 4\lambda^2 \cdot \overline{X}_n\right)$$

which is,

$$\frac{\sqrt{n}(\overline{X}_n^2 - \lambda^2)}{2\lambda\sqrt{\overline{X}_n}} \xrightarrow{d} N(0, 1).$$

use our large sample theory, we have $\overline{X}_n \xrightarrow{P} \lambda$, so we replace the λ in the denominator by $\overline{X}_n$, and we get the C.I as

$$\left(\overline{X}_n^2 - z_{\alpha/2} \cdot 2\overline{X}_n \sqrt{\frac{\overline{X}_n}{n}} \quad , \quad \overline{X}_n^2 + z_{\alpha/2} \cdot 2\overline{X}_n \sqrt{\frac{\overline{X}_n}{n}}\right).$$

Similarly, in $e^{-\lambda}$, we have

$$\frac{\sqrt{n}(e^{-\overline{X}_n} - e^{-\lambda})}{\lambda e^{-\lambda} \cdot \sqrt{e^{-\overline{X}_n}}} \xrightarrow{d} N(0, 1),$$

and the C.I is

$$\left(e^{-\overline{X}_n} - z_{\alpha/2} \cdot 2e^{-\overline{X}_n} \cdot e^{-e^{\overline{X}_n}} \cdot \sqrt{\frac{e^{-\overline{X}_n}}{n}} \ , \ e^{-\overline{X}_n} + z_{\alpha/2} \cdot 2e^{-\overline{X}_n} \cdot e^{-e^{\overline{X}_n}} \cdot \sqrt{\frac{e^{-\overline{X}_n}}{n}}\right).$$

3.3 Bayes estimator

The idea of Bayesian estimation is to work with conditional probability, and we treat the parameter θ as another random variable having its owning distribution, called the **prior distribution** of θ, so in this case $p_\theta(\boldsymbol{x})$ is the conditional distribution given θ, usually denote as $p(\boldsymbol{x}|\theta)$.

Definition 29. Let the random sample $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} f(x;\theta)$ with joint density $p_\theta(\boldsymbol{x})$, assume the prior distribution of θ is given by $\pi(\theta)$, then the **posterior distribution** of θ given data $\boldsymbol{x}$ is defined by

$$\pi(\theta|\boldsymbol{x}) = \frac{p_\theta(\boldsymbol{x}) \cdot \pi(\theta)}{\int_\Theta p_\theta(\boldsymbol{x}) \cdot \pi(\theta) d\theta}.$$

The prior distribution $\pi(\theta)$ reflects our original belief about θ, it is pre-determined. Then the posterior distribution $\pi(\theta|\boldsymbol{x})$ can be viewed as an update of our belief about θ given the observed sample using conditional probability.

Theorem 34. Under square error loss, the Bayesian estimator of θ is simply the mean of the posterior distribution.

The proof of this theorem is given in the next section.

Example 10. Consider the random sample $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} \text{Unif}(0, \theta)$, and the prior distribution for θ is given by

$$\pi(\theta) = \frac{\alpha\beta^\alpha}{\theta^{\alpha+1}}, \text{ where } \alpha, \beta > 0, \theta > 1$$

Find the Bayes estimator of θ under the squared error loss.

Recall that the posterior distribution takes the form

$$\pi(\theta|x) = \frac{p(x|\theta)\pi(\theta)}{\int_\Theta p(x|\theta)\pi(\theta) d\theta},$$

where

$$\begin{aligned} p(x|\theta)\pi(\theta) &= \left(\frac{1}{\theta}\right)^n \cdot \mathbb{1}_{X_{(1)}>0}\mathbb{1}_{X_{(n)}<\theta} \cdot \frac{\alpha\beta^\alpha}{\theta^{\alpha+1}} \cdot \mathbb{1}_{\theta>1} \\ &= \theta^{-(n+\alpha+1)} \cdot \alpha\beta^\alpha \cdot \mathbb{1}_{\theta>\max\{1,X_{(n)}\}} \cdot \mathbb{1}_{X_{(1)}>0}. \end{aligned}$$

So it follows that

$$\begin{aligned} \int_\Theta p(x|\theta)\pi(\theta) d\theta &= \int_{T(\boldsymbol{X})}^\infty \theta^{-(n+\alpha+1)} \alpha\beta^\alpha \mathbb{1}_{X_{(1)}>0} d\theta \text{ where } T(\boldsymbol{X}) = \max\{1, X_{(n)}\} \\ &= \mathbb{1}_{X_{(1)}>0}\alpha\beta^\alpha \left[-\frac{1}{n+\alpha}\theta^{-(n+\alpha)}\right]\Bigg|_{\theta=T(\boldsymbol{X})}^\infty \\ &= \mathbb{1}_{X_{(1)}>0}\alpha\beta^\alpha \left[\frac{1}{n+\alpha}(T(\boldsymbol{X}))^{-(n+\alpha)}\right]. \end{aligned}$$

Hence

$$\pi(\theta|x) = (n+\alpha) \cdot \left(\frac{1}{\theta}\right)^{n+\alpha+1} \cdot [T(\boldsymbol{X})]^{-(n+\alpha)}; \quad \theta > T(\boldsymbol{X}).$$

Now, under the squared error loss, the Bayes estimator of θ is given by the expected value of the posterior distribution, hence

$$\begin{aligned}
\widehat{\theta}_{\text{Bayes}} = \mathbb{E}[\pi(\theta|x)] &= \int_{T(\boldsymbol{X})}^{\infty} \theta \cdot (n+\alpha) \cdot \theta^{-(n+\alpha+1)} \cdot [T(\boldsymbol{X})]^{-(n+\alpha)} d\theta \\
&= (n+\alpha) \cdot [T(\boldsymbol{X})]^{-(n+\alpha)} \int_{T(\boldsymbol{X})}^{\infty} \theta^{-(n+\alpha)} d\theta \\
&= \frac{n+\alpha}{-(n+\alpha)+1} [T(\boldsymbol{X})]^{-(n+\alpha)} [\theta^{-(n+\alpha)+1}] \Big|_{\theta=T(\boldsymbol{X})}^{\infty} \\
&= \frac{n+\alpha}{n+\alpha-1} T(\boldsymbol{X}).
\end{aligned}$$

where $T(\boldsymbol{X}) = \max\{1, X_{(n)}\}$.

Example 11. Consider the random sample $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} \text{Bern}(\theta)$, and the prior distribution of θ is $\pi(\theta) \sim \text{Beta}(\alpha, \beta), \alpha, \beta > 0$. Under square error loss, find the Bayesian estimator of θ. The Beta distribution has density

$$\pi(\theta) \sim \text{Beta}(\alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1}.$$

Note that in the formula

$$\pi(\theta|x) = \frac{p(x|\theta)\pi(\theta)}{\displaystyle\int_{\Theta} p(x|\theta)\pi(\theta)d\theta},$$

the denominator is independent of θ, hence we have

$$\begin{aligned}
\pi(\theta|\boldsymbol{x}) &\propto p_{\theta}(\boldsymbol{x}) \cdot \pi(\theta) \\
&= \theta^{\sum x_i}(1-\theta)^{n-\sum x_i} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1} \\
&= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{n\overline{x}_n+\alpha-1} \cdot (1-\theta)^{n-n\overline{x}_n+\beta-1}
\end{aligned}$$

which we see it takes the form of another Beta distribution, and if we match the coefficients, we would have the posterior distribution given by

$$\pi(\theta|\boldsymbol{x}) \sim \text{Beta}(n\overline{x}_n + \alpha, n - n\overline{x}_n + \beta).$$

Then under square error loss, the Bayes estimator is just the mean of the posterior distribution, given by

$$\widehat{\delta}(\boldsymbol{X})_{\text{Bayes}} = \mathbb{E}[\pi(\theta|\boldsymbol{x})] = \frac{n\overline{X}_n + \alpha}{n+\alpha+\beta}.$$

Definition 30. Suppose $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} f(x;\theta)$, with a prior imposed: $\pi(\theta)$. Then let $X_{n+1} \sim f(x;\theta)$ be a new sample, the **posterior predictive distribution** of $X_{n+1}|X_1, \cdots, X_n$ is

$$f(x_{n+1}|x_1, \cdots, x_n) = \int p(x_{n+1}|\theta) f(\theta|x_1, \cdots, x_n) d\theta,$$

where $p(x_{n+1}|\theta) = f(x_{n+1};\theta)$ is the likelihood, and $f(\theta|x_1, \cdots, x_n)$ is the posterior distribution of θ.

Theorem 35 (Bernstein-von Mises). Assume $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} P_{\theta^*}$. Let $\widehat{\theta}_n$ be the MLE of θ^* and $\widehat{\sigma} = 1/\sqrt{n\mathcal{I}(\widehat{\theta}_n)}$. Under some regularity conditions, the posterior distribution of θ^*, $p(\theta^*|x_1, \cdots, x_n)$ converges to $N(\widehat{\theta}_n, \widehat{\sigma})$ in total variation (TV) metric, where

$$\mathrm{TV}(p, q) = \sup_{x \in \Omega} |p(x) - q(x)|.$$

Definition 31. Given $X \sim f(x;\theta)$, the **Jeffreys prior** of θ, is given by

$$\pi(\theta) \propto \det(\mathcal{I}(\theta))^{1/2} = \left\{ \prod_{j=1}^{k} \mathbb{E}\left[\left(\frac{\partial}{\partial \theta_j} \log f(x;\theta) \right)^2 \right] \right\}^{1/2} =^* \left(\prod_{j=1}^{k} \left| \mathbb{E}\left[\frac{\partial^2}{\partial \theta_j^2} \log f(x;\theta) \right] \right| \right)^{1/2}.$$

3.4 Statistical decision theory

Definition 32. Let Θ be the parameter space, a **loss function** is a measurable function such that $L(\theta, \widehat{\theta}) : \Theta \times \Theta \to \mathbb{R}$, where θ is the true parameter, and $\widehat{\theta}$ is the estimate (or decision rule).

Common choice of loss function is the square error loss (L^2 loss): $L(\theta, \widehat{\theta}) = (\theta - \widehat{\theta})^2$, or L^1 loss: $L(\theta, \widehat{\theta}) = |\theta - \widehat{\theta}|$.

Definition 33. Given the loss function $L(\theta, \widehat{\theta})$ (θ is the true parameter, $\widehat{\theta}$ is the estimate), the **risk** of the estimate $\widehat{\theta}$ is given by

$$R(\theta, \widehat{\theta}) = \mathbb{E}_\theta\left[L(\theta, \widehat{\theta}) \right] = \int L(\theta, \widehat{\theta}) dF_\theta(x),$$

i.e, expectation of the loss function L with respect to the true parameter θ.

Example 12. Suppose $X_1, \cdots, X_n \overset{\text{i.i.d}}{\sim} \mathrm{Bern}(p)$, and $\widehat{p}_1 = \overline{X}_n$; $\widehat{p}_2 = \mathbb{E}[p|X_1, \cdots, X_n]$ where $p \sim \mathrm{Beta}(\sqrt{n}/2, \sqrt{n}/2)$. Using L^2 loss, find the risk of $\widehat{p}_1, \widehat{p}_2$.

Since $\widehat{p}_1$ is unbiased estimate of p, hence

$$R(p, \widehat{p}_1) = \mathbb{E}_p[(\overline{X}_n - p)^2] = \mathbb{V}\mathbf{ar}(\overline{X}_n) = \frac{p(1-p)}{n}.$$

For $\widehat{p}_2$, we know the posterior of $p|X_1, \cdots, X_n$ is $\mathrm{Beta}(\sqrt{n}/2 + n\overline{X}_n, \sqrt{n}/2 + n - n\overline{X}_n)$, hence

$$\widehat{p}_2 = \frac{\frac{1}{2} + \sqrt{n} \cdot \overline{X}_n}{1 + \sqrt{n}}.$$

Definition 34. Given $R(\theta, \widehat{\theta})$, we define the following estimators based on the risk function:

- The **maximum risk** estimator of θ is given by $\bar{R}(\widehat{\theta}) = \sup_{\theta \in \Theta} R(\theta, \widehat{\theta})$;
- The **minimax** estimator of θ is given by $\widehat{\theta}$ such that $\sup_\theta R(\theta, \widehat{\theta}) = \inf_{\widetilde{\theta}} \sup_\theta R(\theta, \widetilde{\theta})$;

In fact, the following estimator is often more useful and easy to compute:

Definition 35. Given data $\boldsymbol{X} = (X_1, \cdots, X_n) \overset{\text{i.i.d}}{\sim} p(x; \theta)$, the loss function $L(\delta(\boldsymbol{X}), \theta)$ measures the loss when θ is estimated by $\delta(\boldsymbol{X})$. Suppose $\theta \sim \pi$ is the prior imposed on θ, the Bayes' risk of $\delta(\boldsymbol{X})$ is defined by

$$R(\delta(\boldsymbol{X})) = \mathbb{E}_\pi \Big[\mathbb{E}_{x|\theta}[L(\delta(\boldsymbol{X}), \theta)] \Big],$$

where the Bayes estimator is given by $\widehat{\delta(\boldsymbol{X})}_{\text{Bayes}} = \arg\min_{\delta(\boldsymbol{X})} R(\delta)$.

By definition, we may compute $R(\delta(\boldsymbol{X}))$ (where we assume certain regularity conditions hold):

$$\begin{aligned} R(\delta(\boldsymbol{X})) &= \int_\Theta \int_X L(\delta(\boldsymbol{X}), \theta) p_\theta(\boldsymbol{x}) d\boldsymbol{x} \pi(\theta) d\theta \\ &= \int_X \int_\Theta L(\delta(\boldsymbol{X}), \theta) \pi(\theta|\boldsymbol{x}) d\theta p_\theta(\boldsymbol{x}) dx \\ &= \int_\Theta L(\delta(\boldsymbol{X}), \theta) \pi(\theta|\boldsymbol{x}) d\theta \end{aligned}$$

One common loss function is squared-error loss (mean-square error): $L(\delta(\boldsymbol{X}) - \theta) = (\delta(\boldsymbol{X}) - \theta)^2$, using this loss function, we have

$$\widehat{\delta(\boldsymbol{X})}_{\text{Bayes}} = \arg\min_{\delta(\boldsymbol{X})} \left\{ \int_\Theta (\delta(\boldsymbol{X}) - \theta)^2 \pi(\theta|\boldsymbol{x}) d\theta \right\}.$$

In order to find the $\delta(\boldsymbol{X})$ that minimizes the term above, we differentiate with respect to $\delta(\boldsymbol{X})$ and we have

$$\frac{df}{d\delta(\boldsymbol{X})} = \int_\Theta 2(\delta(\boldsymbol{X}) - \theta) \pi(\theta|\boldsymbol{x}) d\theta.$$

Set the derivative equal to 0 and we solve for $\delta(\boldsymbol{X})$:

$$\delta(\boldsymbol{X}) \int_\Theta \pi(\theta|\boldsymbol{x}) d\theta = \int_\Theta \theta \pi(\theta|\boldsymbol{x}) d\theta.$$

Note that the integral on the left hand side is equal to 1, and hence we have

$$\delta(\boldsymbol{X}) = \int_\Theta \theta \pi(\theta|\boldsymbol{x}) d\theta = \mathbb{E}[\theta|\boldsymbol{X} = \boldsymbol{x}].$$

The above demonstration is exactly the proof of Theorem 34: Under square error loss, the Bayes estimator of θ is simply the mean of the posterior distribution.

Theorem 36. Let $\widehat{\theta}^\pi$ be the Bayes estimator with prior π, so $r(\pi, \widehat{\theta}^\pi) = \inf_{\widetilde{\theta}} r(\pi, \widetilde{\theta})$. Suppose $R(\theta, \widehat{\theta}^\pi) \leq r(\pi, \widehat{\theta}^\pi)$ for all $\theta \in \Theta$, then $\widehat{\theta}^\pi$ is the minimax.

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